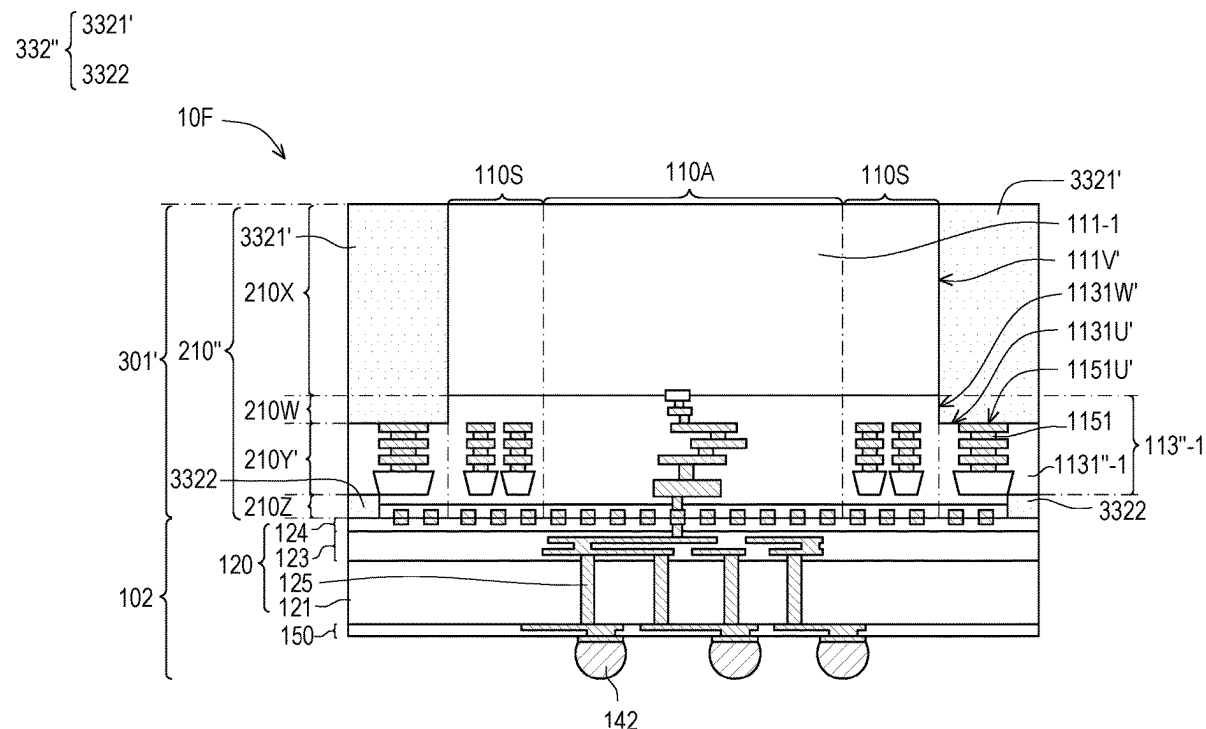


(43) **Pub. Date:** **Sep. 11, 2025**



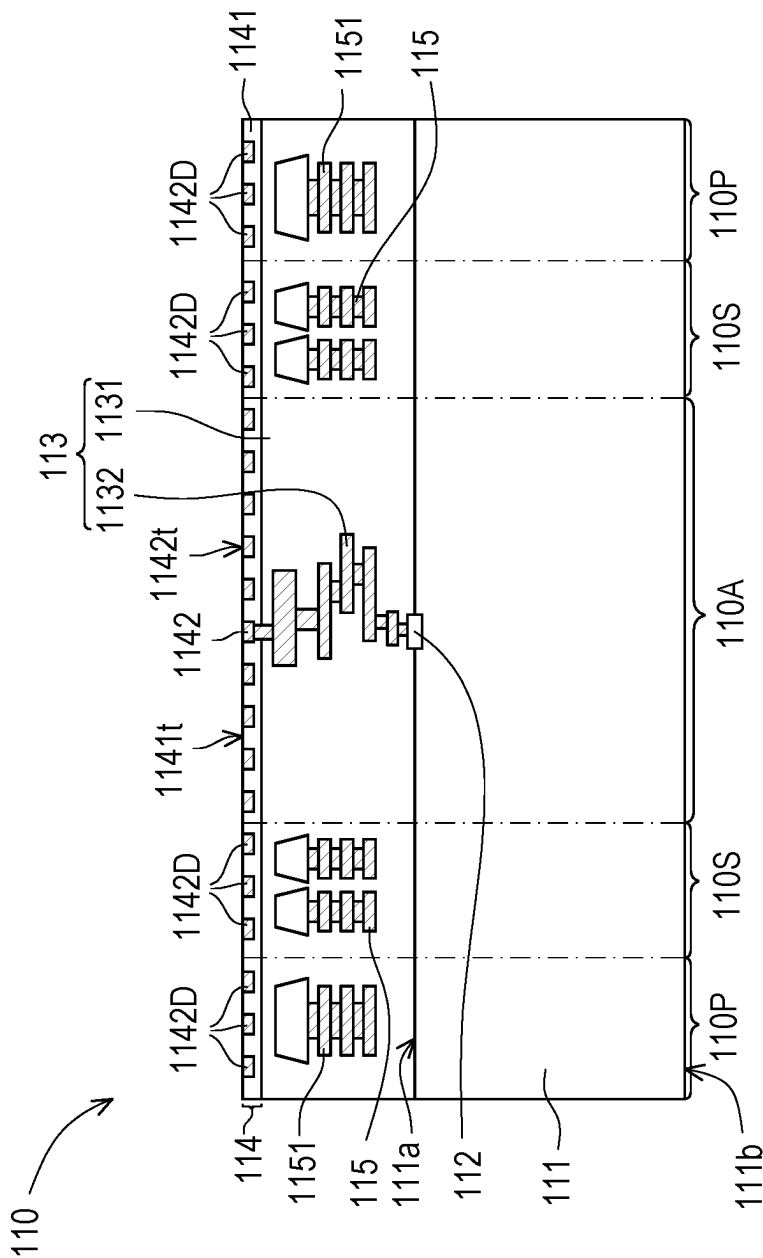
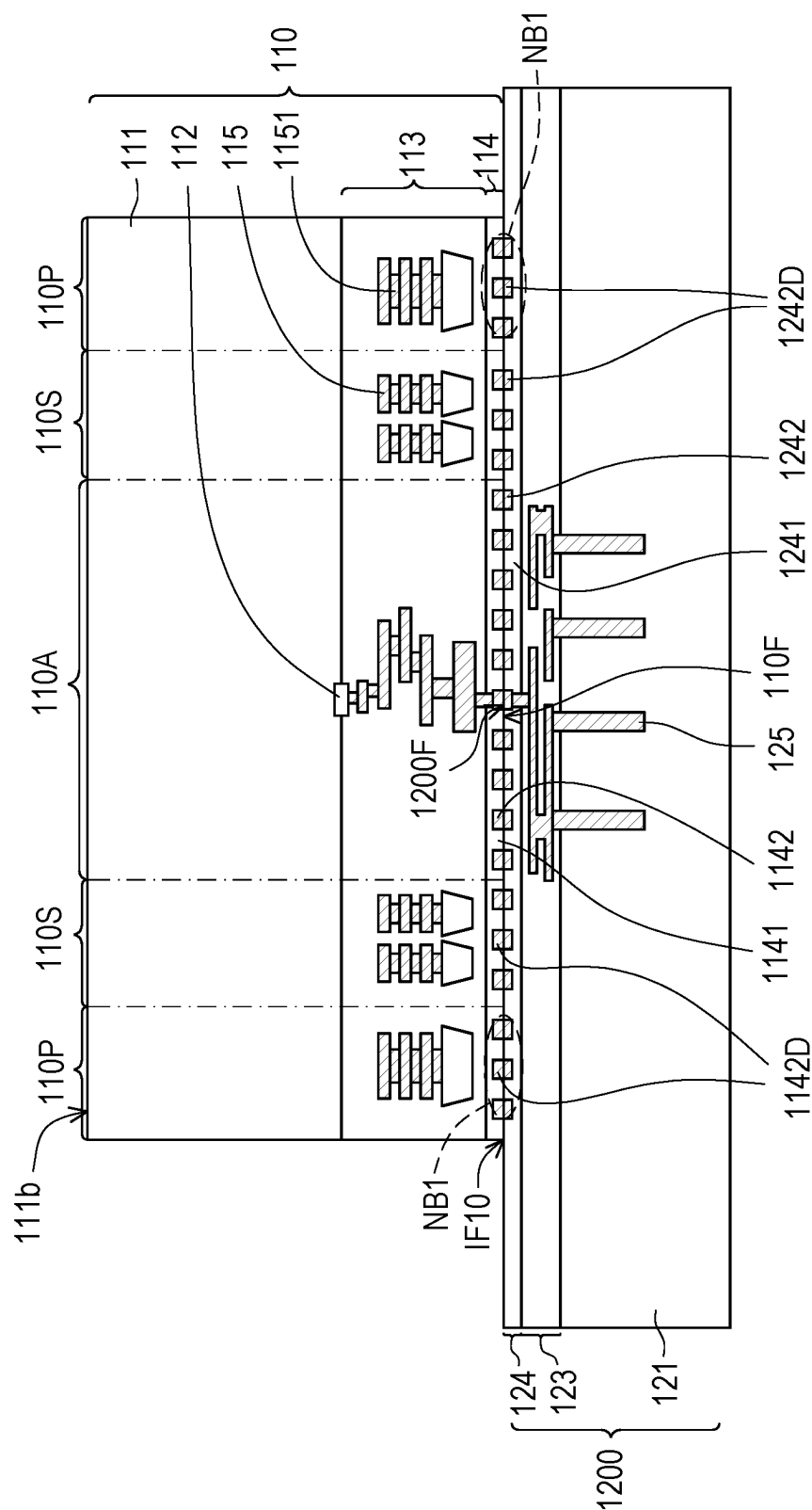


FIG. 1A



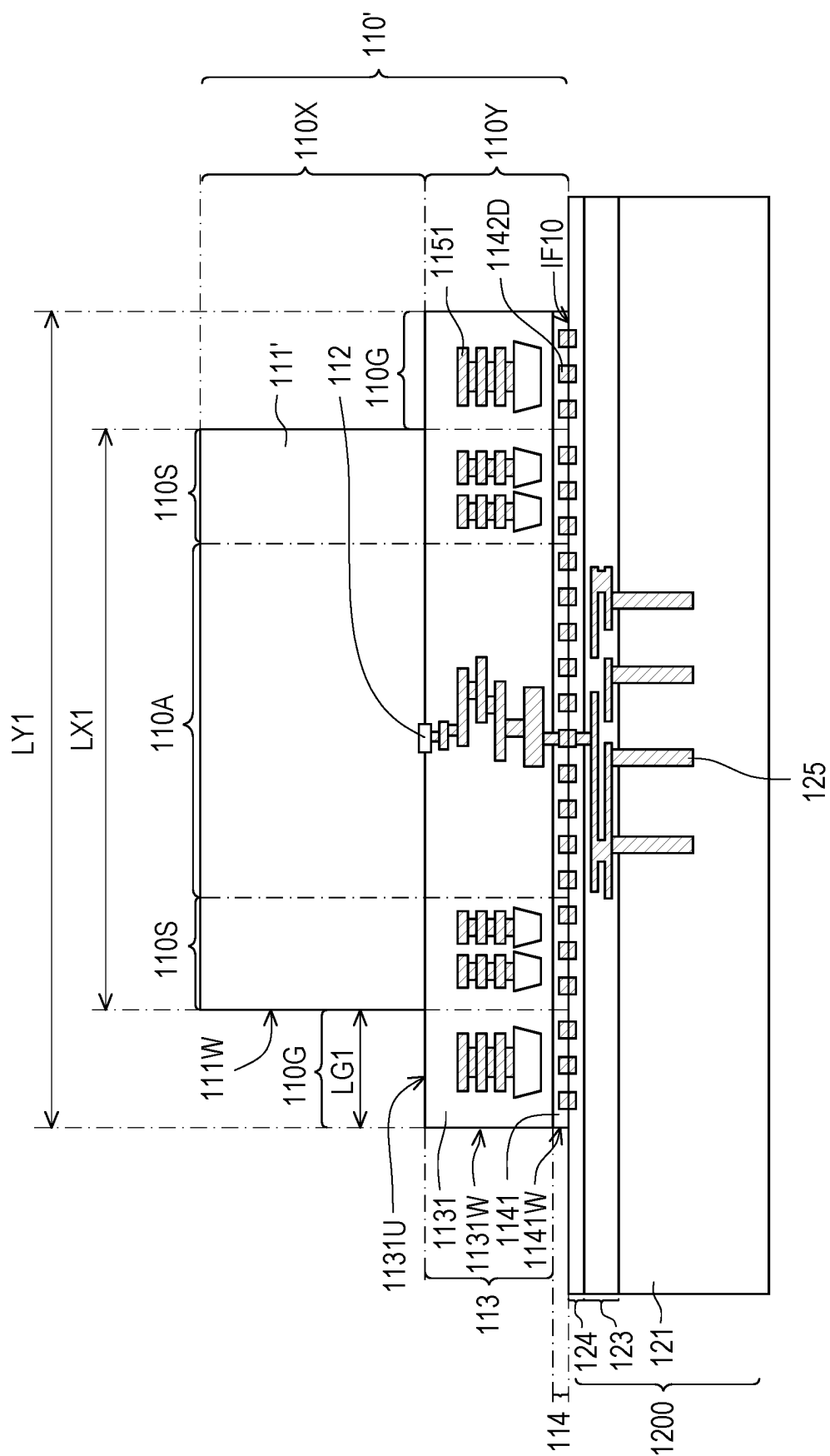


FIG. 2B

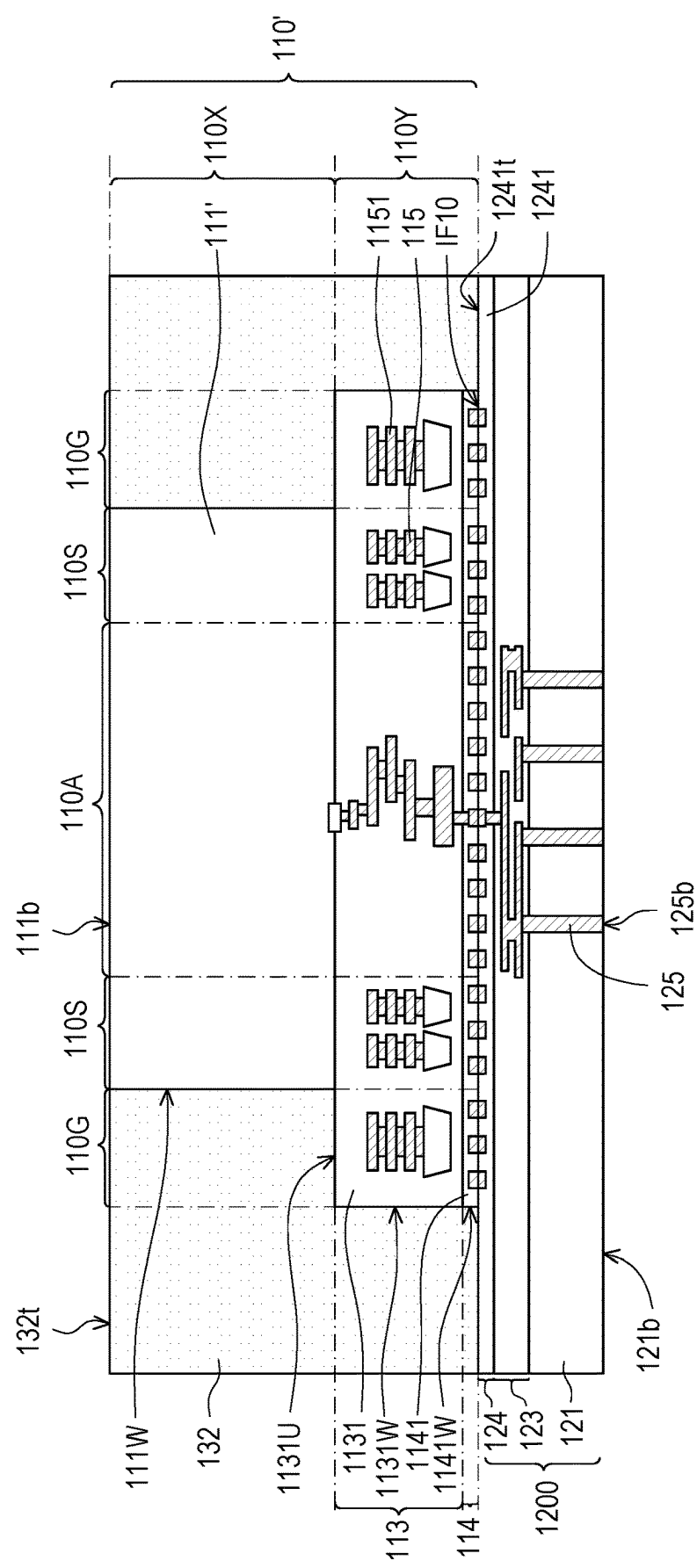


FIG. 2C

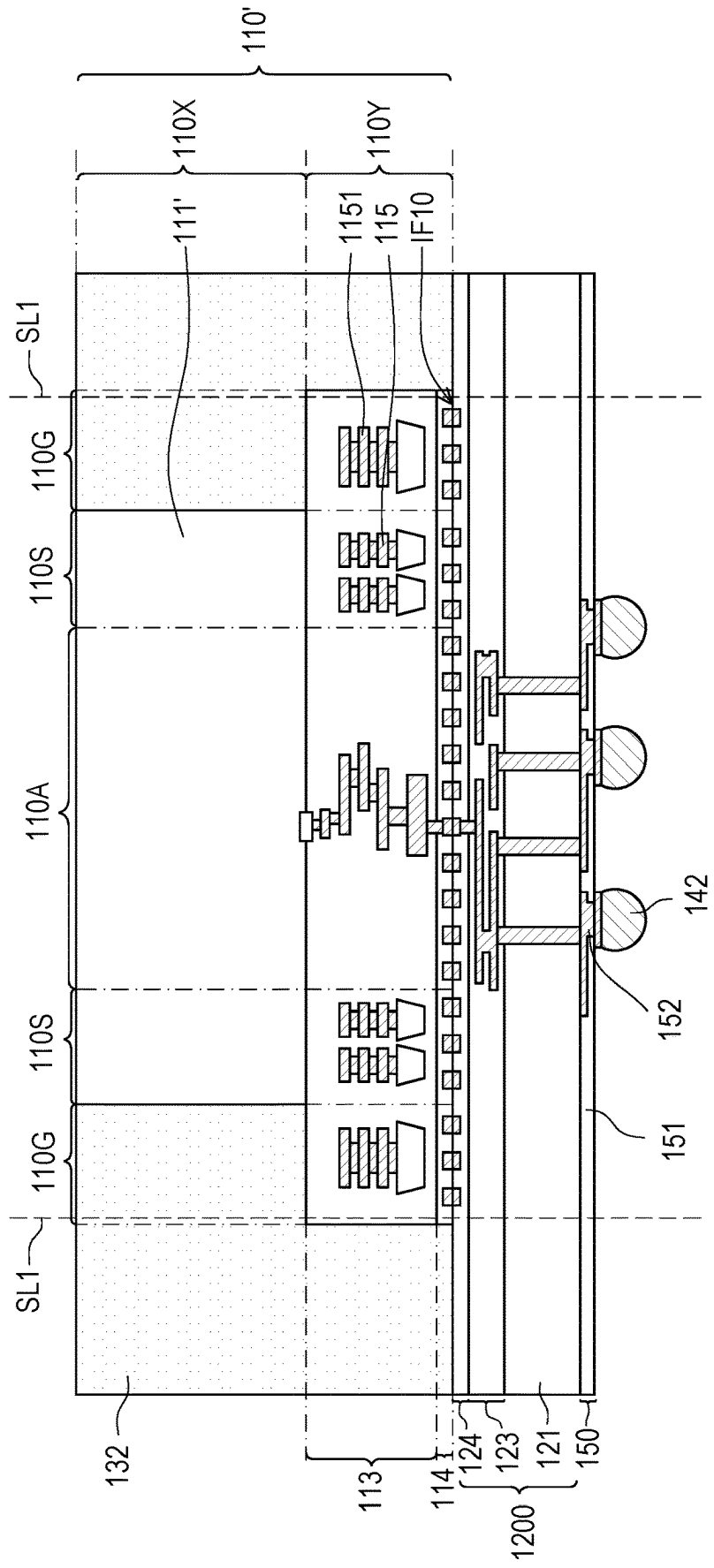


FIG. 2D

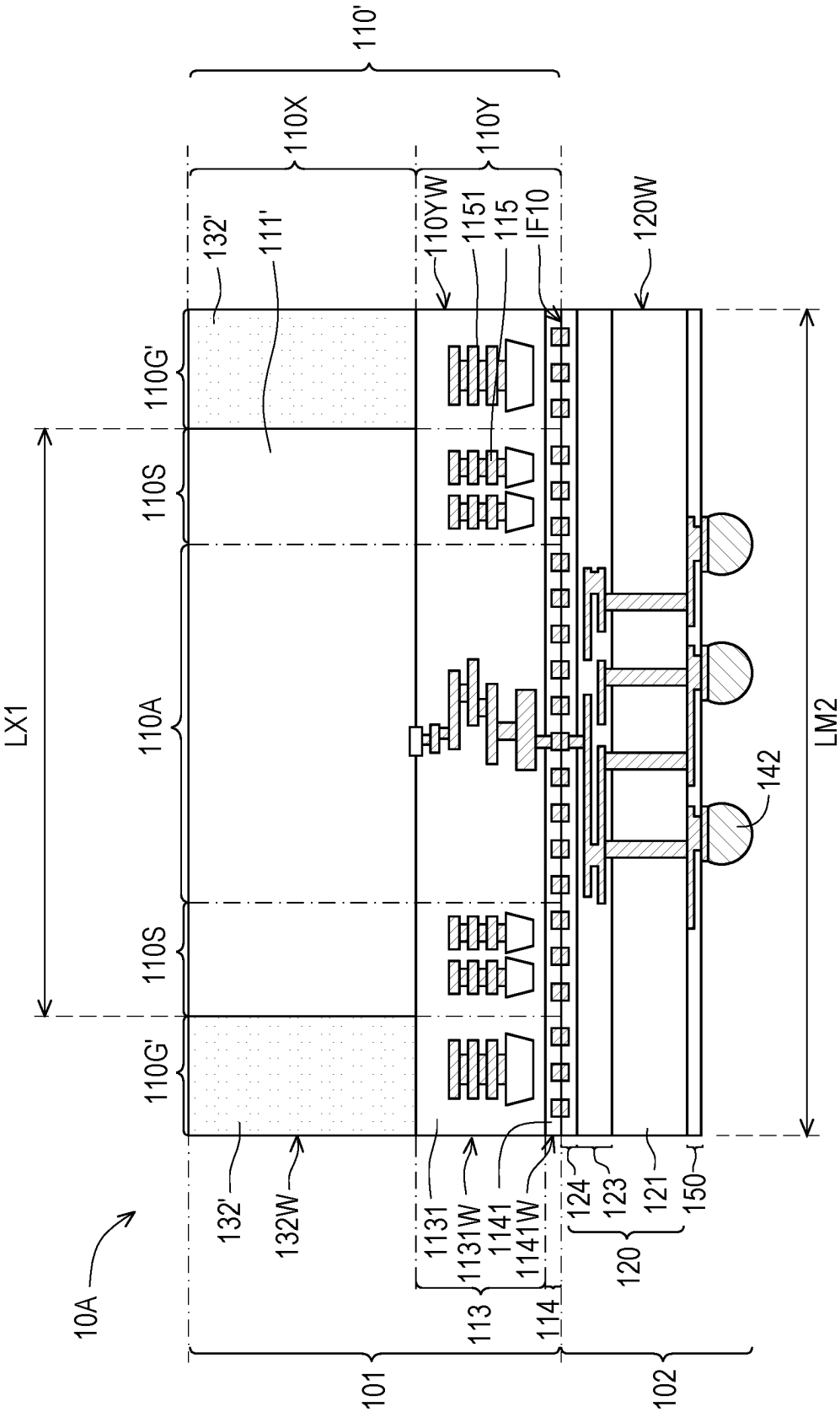


FIG. 2E

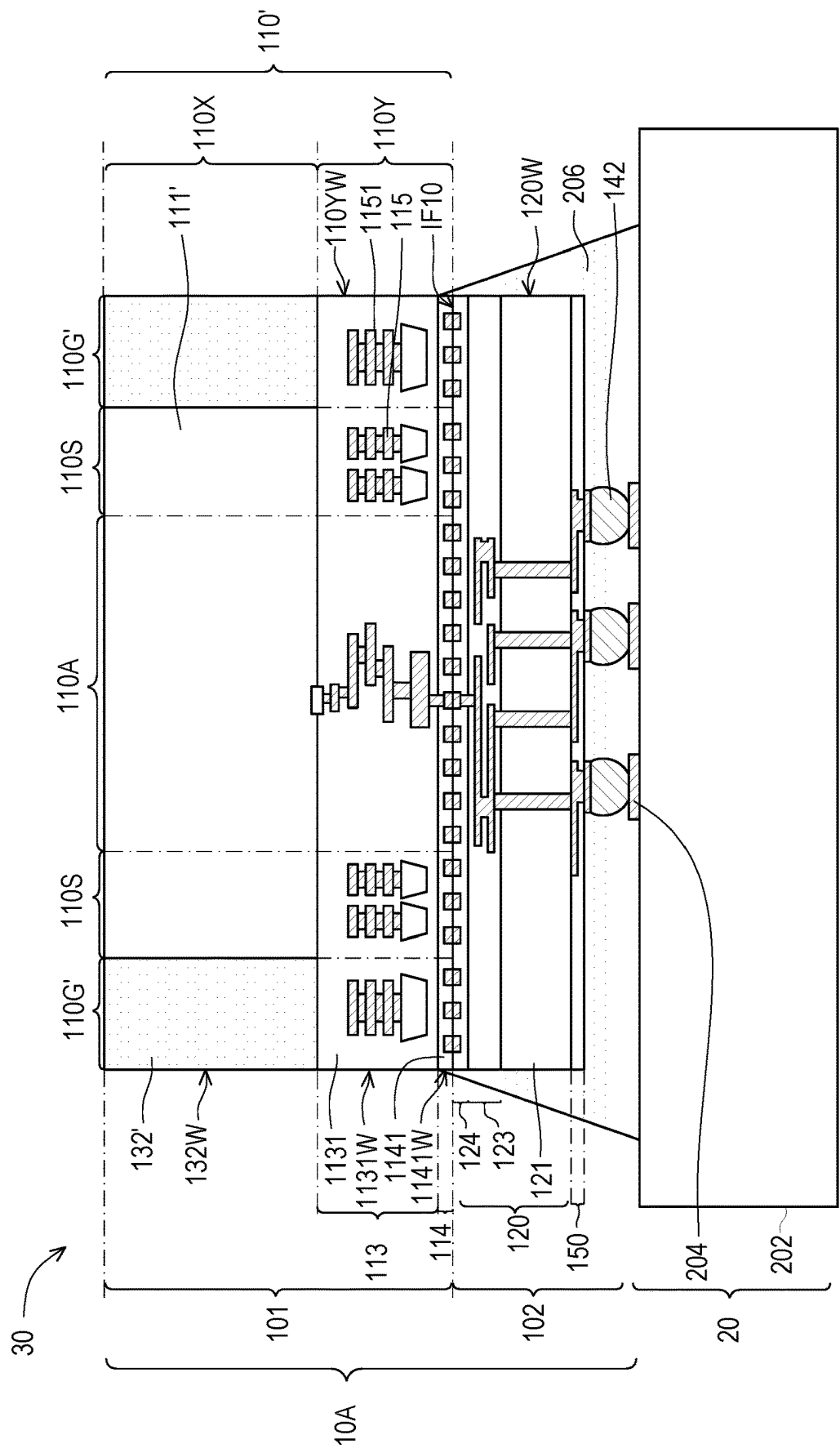


FIG. 2F

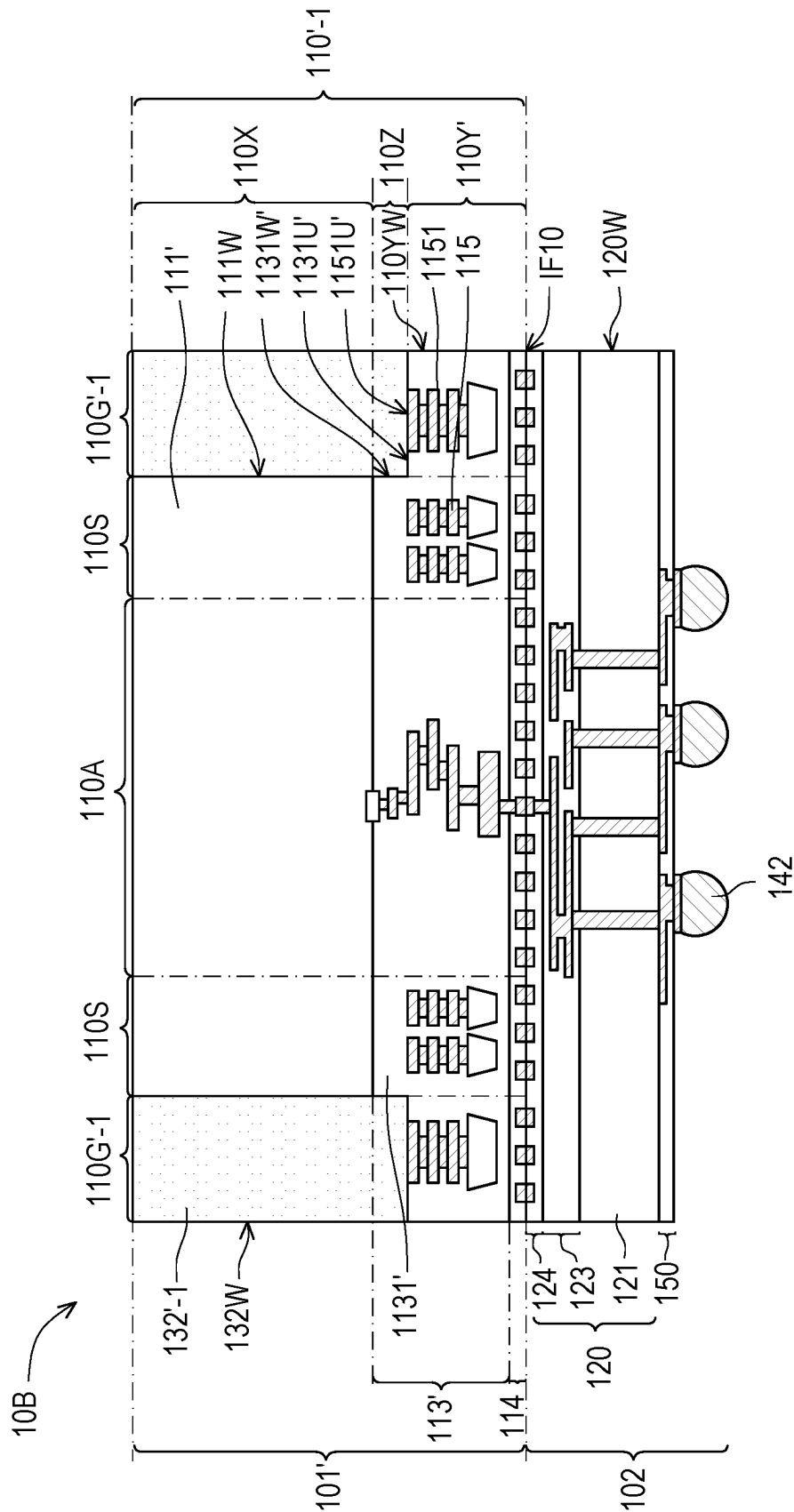


FIG. 3A

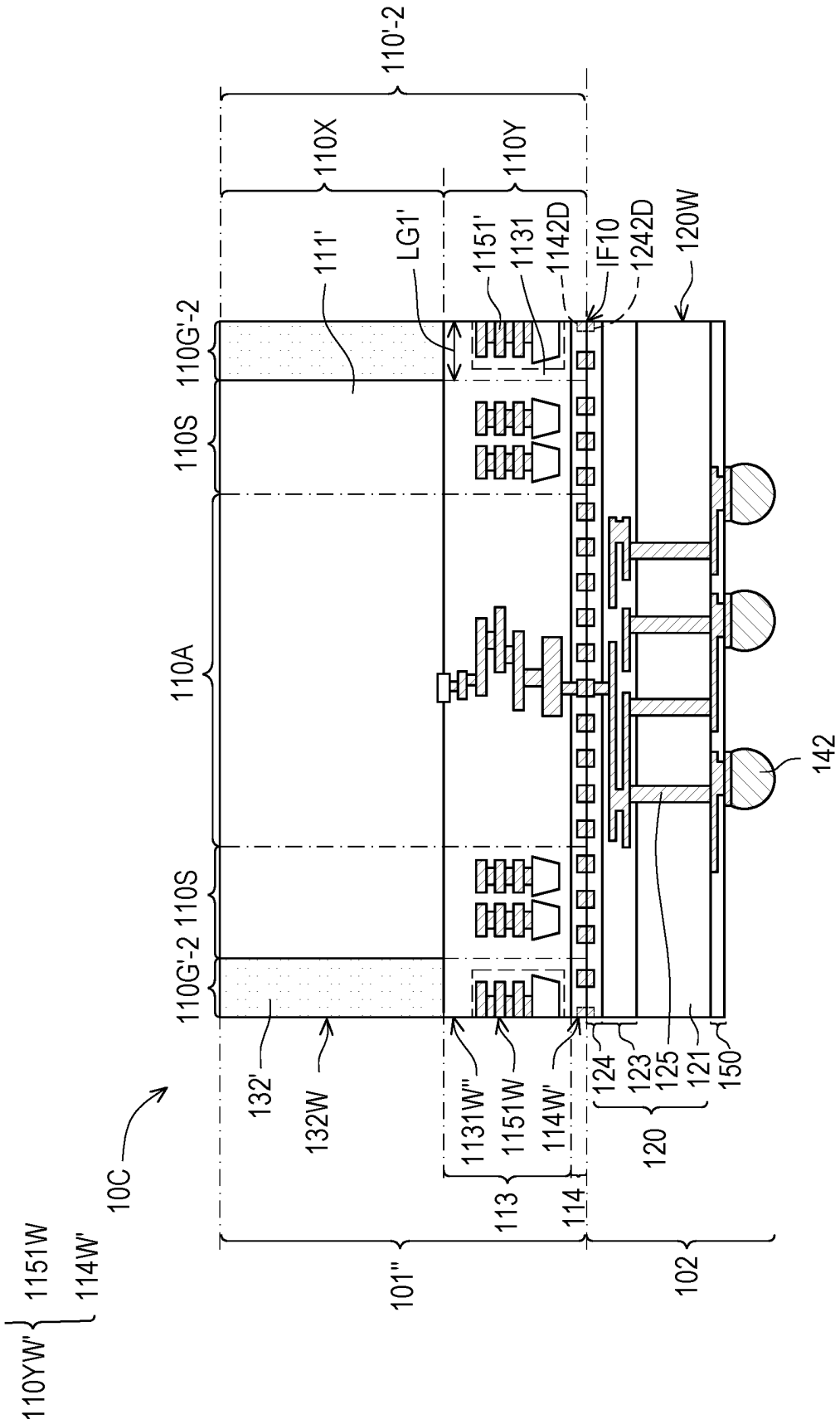
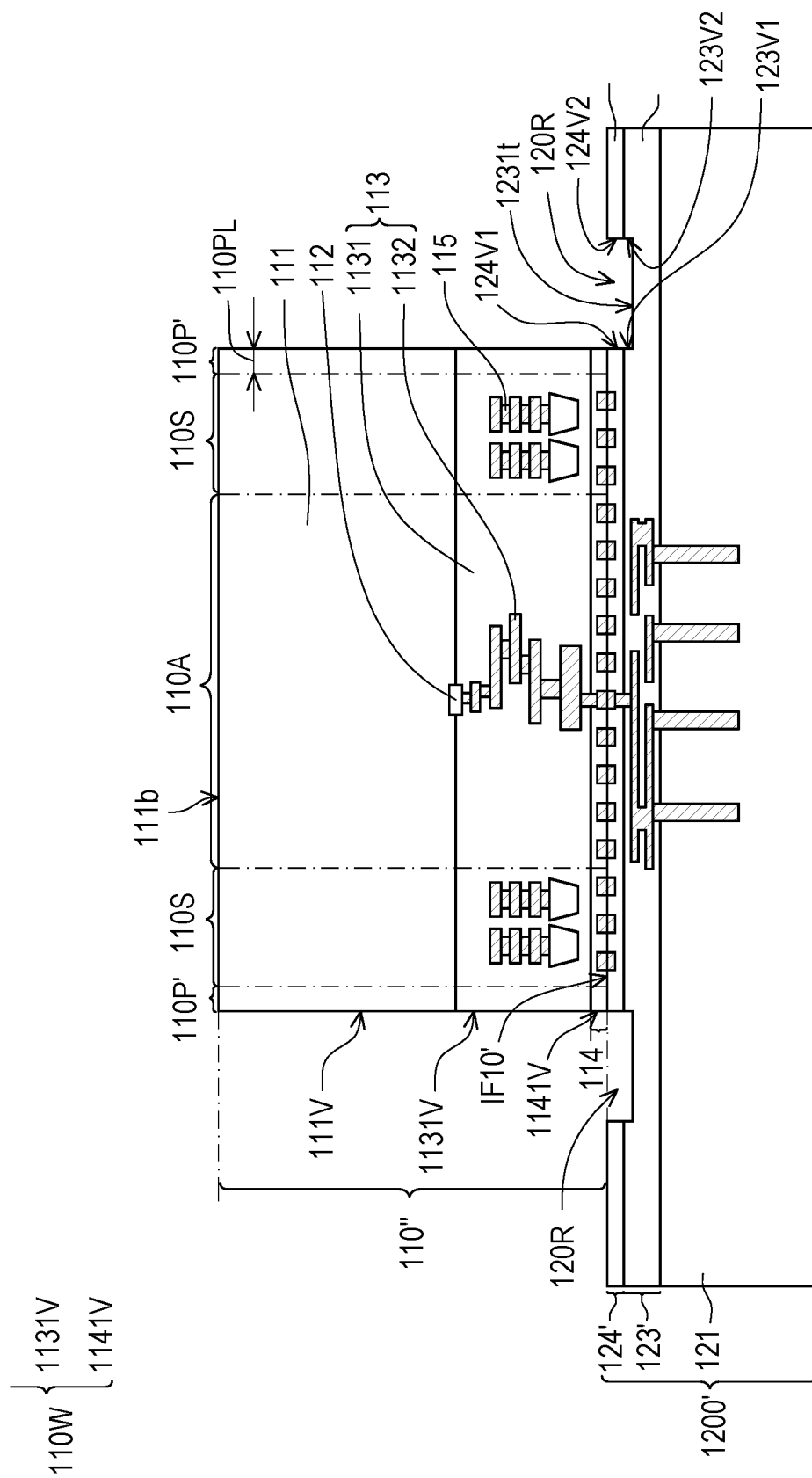


FIG. 3B



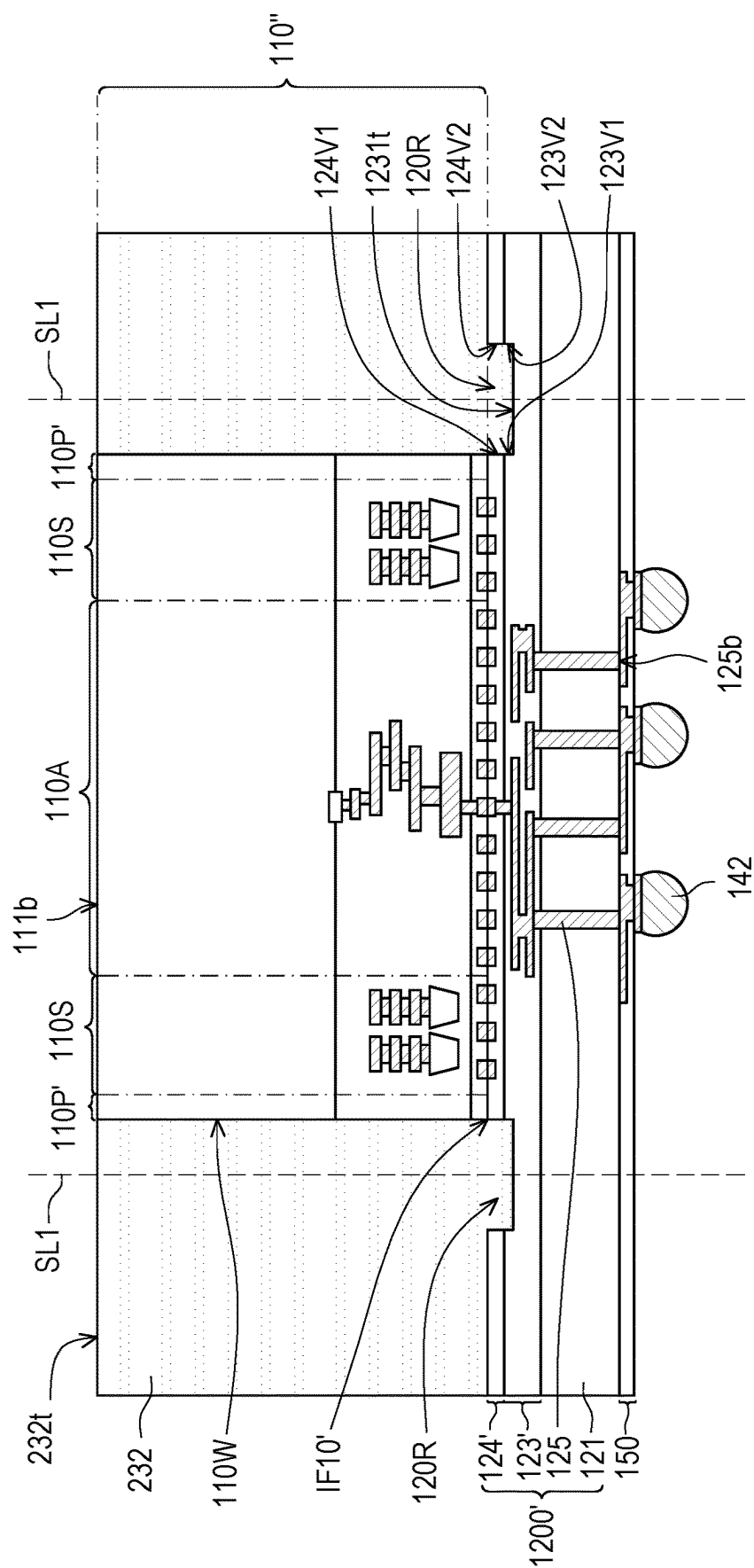


FIG. 4B

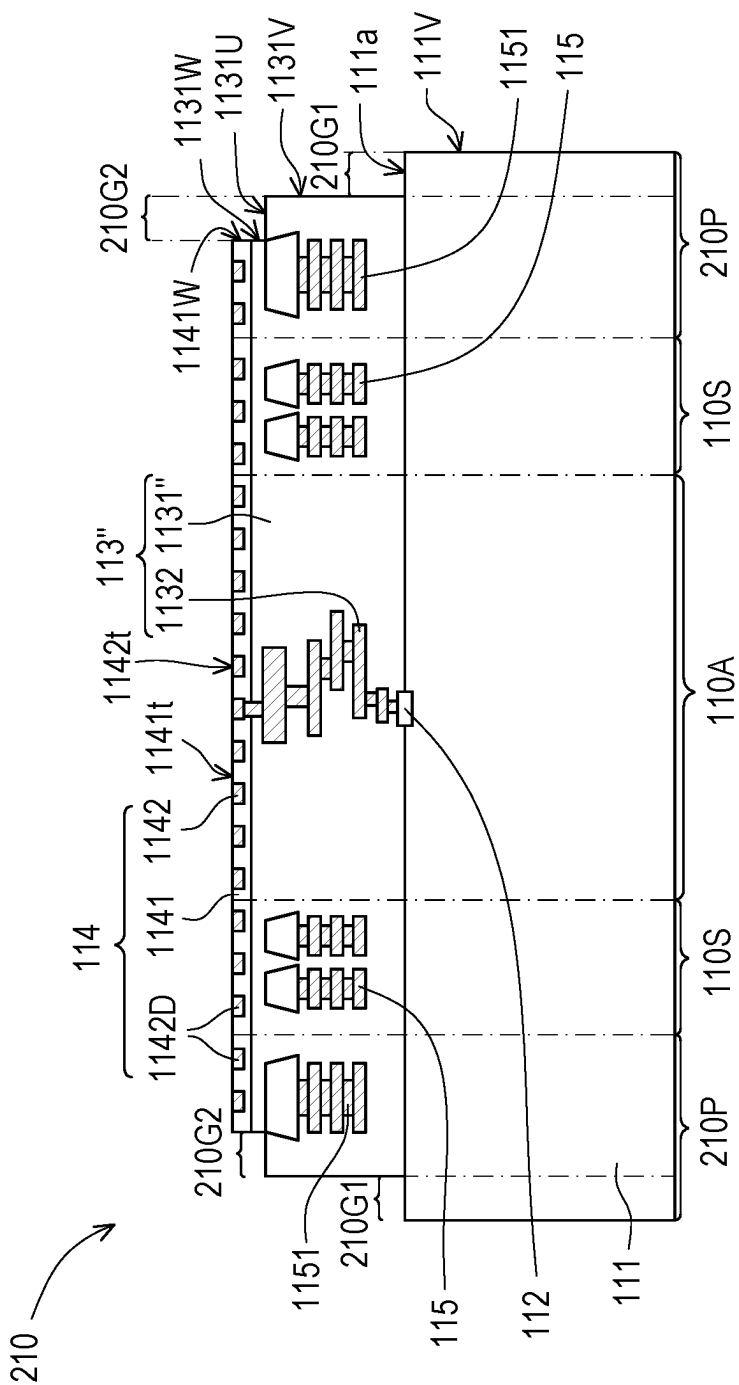


FIG. 5A

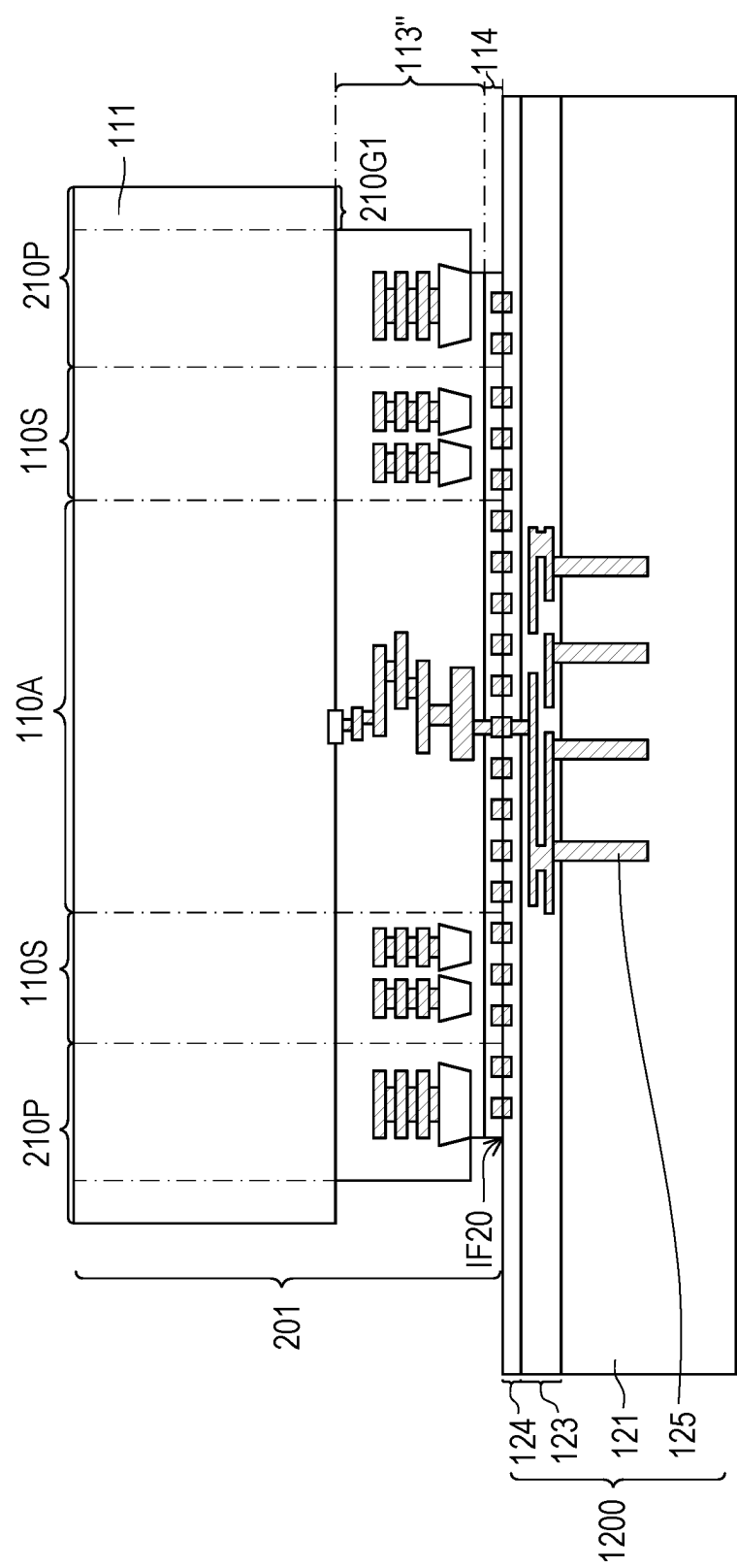


FIG. 5B

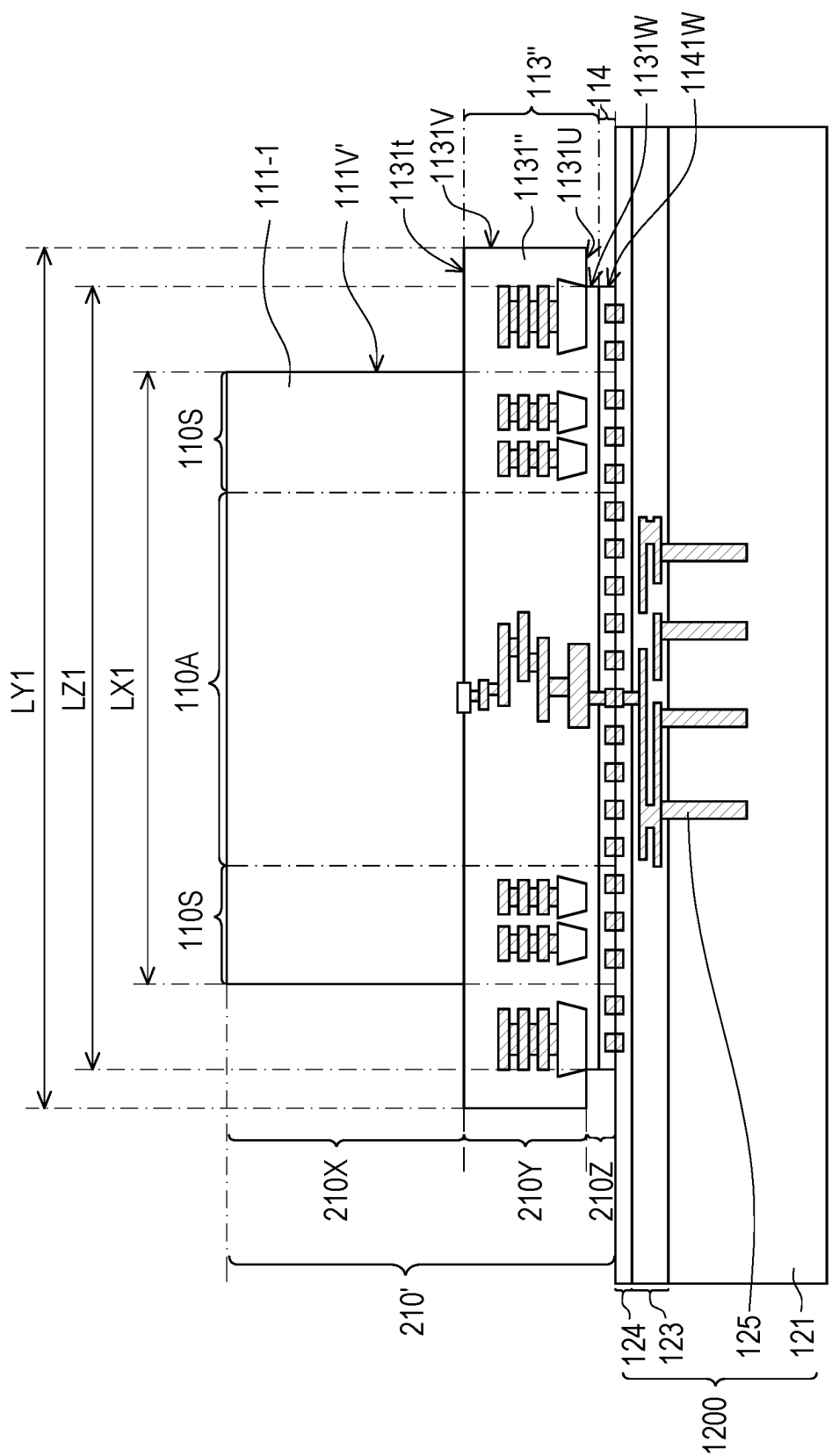


FIG. 5C

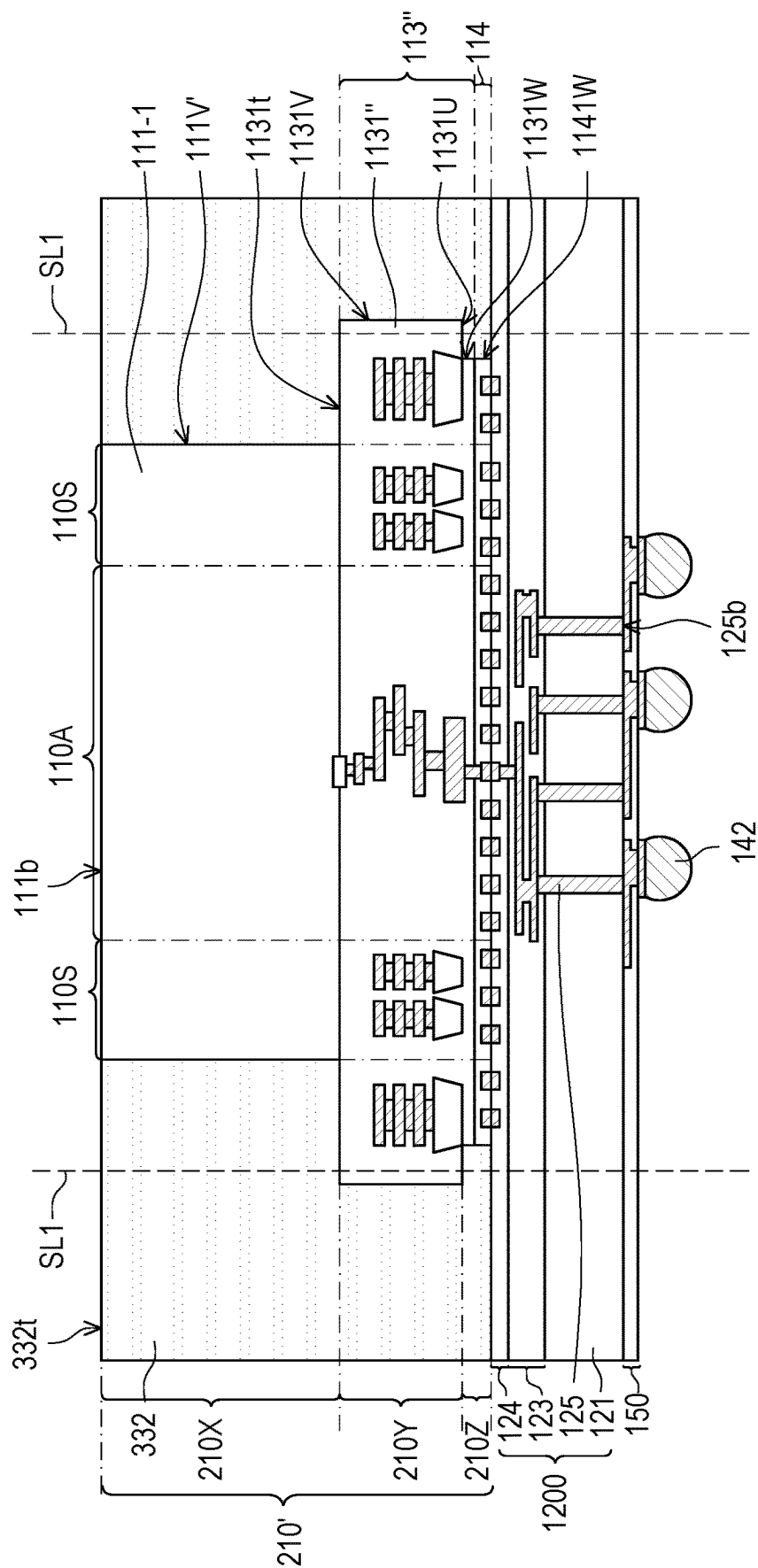


FIG. 5D

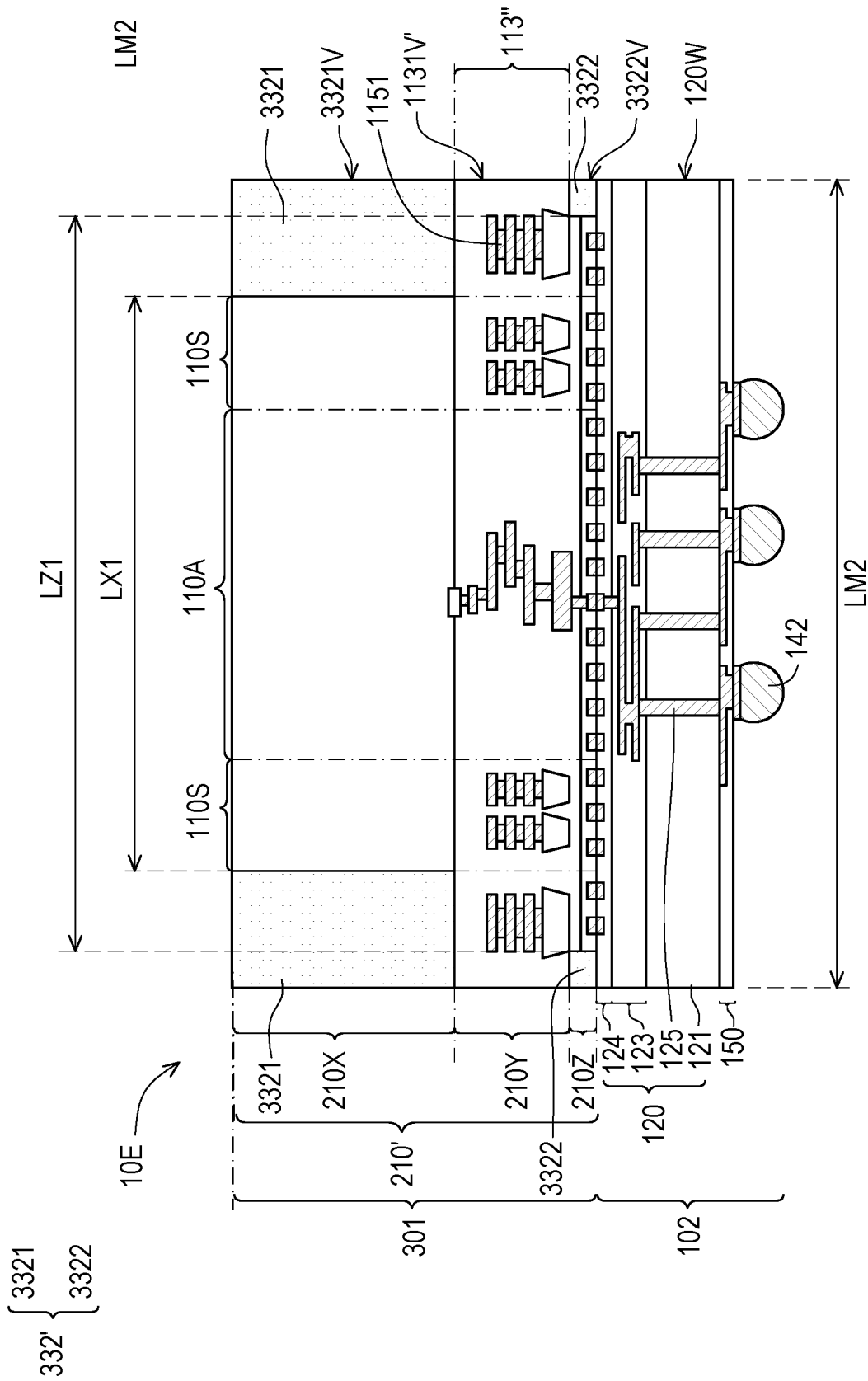
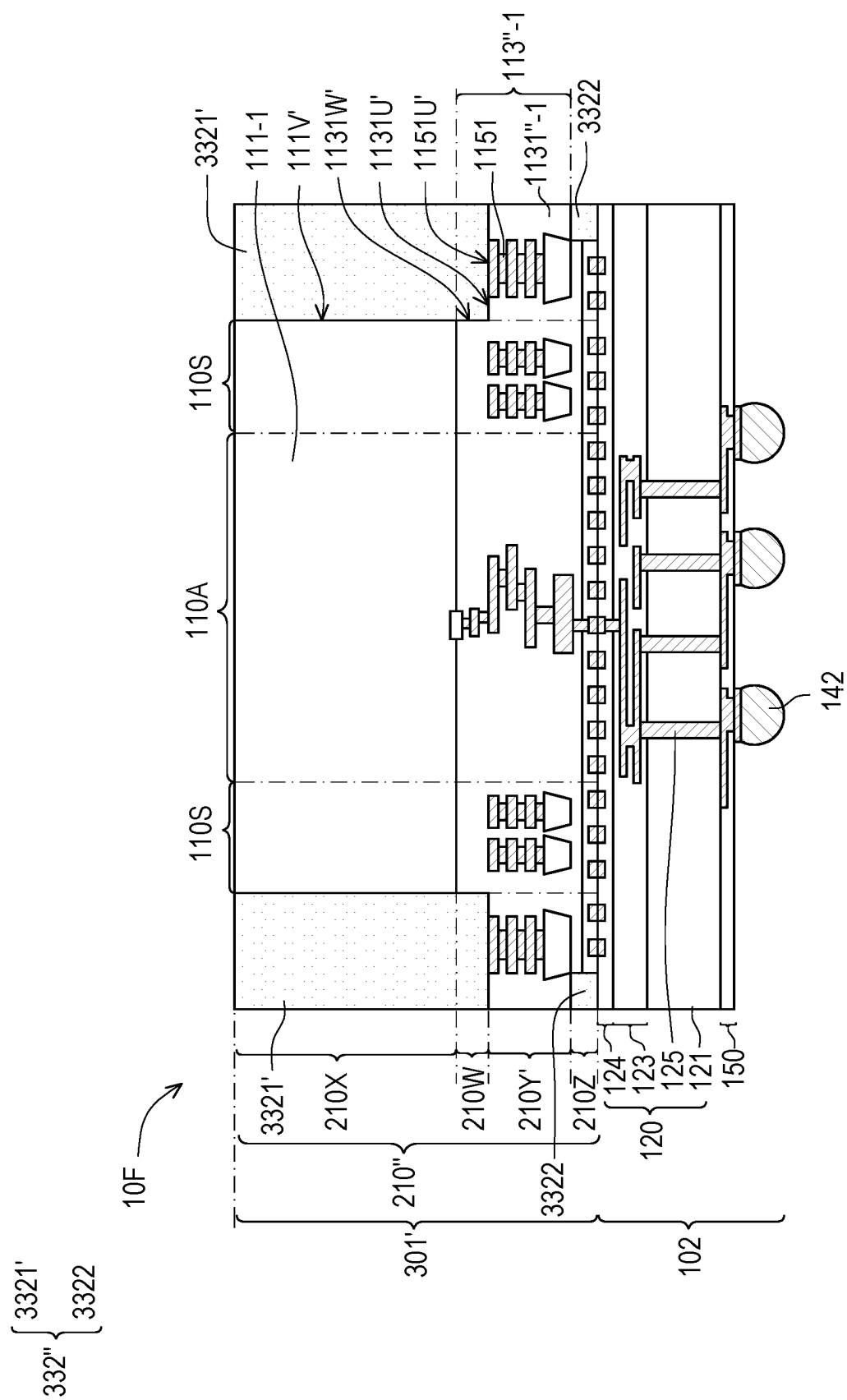


FIG. 5E



SEMICONDUCTOR STRUCTURE AND MANUFACTURING METHOD THEREOF

BACKGROUND

[0001] The semiconductor industry has experienced rapid growth due to continuous improvement in integration density of various components, e.g., transistors, diodes, resistors, capacitors, etc. For the most part, this improvement in integration density has come from successive reductions in minimum feature size, which allows more components to be integrated into a given area. Technological advances in integrated circuit (IC) design have produced generations of ICs where each generation has smaller and more complex circuit designs than the previous generation. There is continuous effort in developing new mechanisms of forming semiconductor structures having improved electrical performance.

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIG. 1A illustrates a schematic cross-sectional view of a first semiconductor die, in accordance with some embodiments.

[0004] FIG. 1B illustrates a schematic cross-sectional view of a semiconductor wafer, in accordance with some embodiments.

[0005] FIGS. 2A-2F illustrate schematic cross-sectional views of intermediate steps during a process for forming an integrated circuit package including a semiconductor structure, in accordance with some embodiments.

[0006] FIG. 3A and FIG. 3B illustrate schematic cross-sectional views of variations of a semiconductor structure, in accordance with some embodiments.

[0007] FIGS. 4A-4C illustrate schematic cross-sectional views of intermediate steps during a process for forming a semiconductor structure, in accordance with some embodiments.

[0008] FIG. 5A illustrates a schematic cross-sectional view of a first semiconductor die, in accordance with some embodiments.

[0009] FIGS. 5B-5E illustrate schematic cross-sectional views of intermediate steps during a process for forming a semiconductor structure, in accordance with some embodiments.

[0010] FIG. 6 illustrates schematic cross-sectional views of a semiconductor structure, in accordance with some embodiments.

DETAILED DESCRIPTION

[0011] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in

which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0012] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0013] Embodiments discussed herein are to provide various semiconductor structures and methods for forming the same. For example, a semiconductor structure is formed by bonding a first semiconductor die to a second semiconductor die and forming an insulating encapsulant over the second semiconductor die to surround the first semiconductor die. The internal stress originates from a difference in thermal expansion among the first and second semiconductor dies and the insulating encapsulant. The thermal expansion difference is because a difference in a coefficient of thermal expansion (CTE) of the materials among the first and second semiconductor dies and the insulating encapsulant. In addition, the large CTE mismatch among the insulating encapsulant and the first and second semiconductor dies generates the stress in the semiconductor structure, especially at the bonding interface of the first and second semiconductor dies. Delamination in the bonded structure may occur or become worse during the formation of the insulating encapsulant. For example, the delamination propagates from a non-functional (or peripheral) region of the bonded structure toward a functional (or central) region of the bonded structure and such propagation may cause device failure.

[0014] According to some embodiments, a portion of the first semiconductor die corresponding to the non-functional (or peripheral) region of the bonded structure is removed prior to the formation of the insulating encapsulant. This may help to reduce the risk of delamination propagation during the formation of the insulating encapsulant. According to some embodiments, a portion of the bonded structure with non-bond areas is removed prior to the formation of the insulating encapsulant. In this manner, the likelihood of delamination propagation is eliminated. A semiconductor structure with reduced defects, improved reliability, and improved yield may be achieved. Accordingly, various embodiments provide semiconductor structures with reduced stress and improved bonding integrity.

[0015] FIG. 1A illustrates a schematic cross-sectional view of a first semiconductor die, in accordance with some embodiments. It should be noted that FIG. 1A is provided for illustrative purposes only, and the first semiconductor die may utilize fewer or additional elements according to some embodiments. Referring to FIG. 1A, a first semiconductor die 110 may be provided. The first semiconductor die 110 may be formed in a wafer (not shown), which may include

different die regions that are singulated in subsequent steps to form a plurality of first semiconductor dies **110**. The first semiconductor die **110** may be a logic device (e.g., central processing unit (CPU), graphics processing unit (GPU), microcontroller, etc.), a memory device (e.g., dynamic random access memory (DRAM) die, static random access memory (SRAM) die, etc.), a power management device (e.g., power management integrated circuit (PMIC) die), a radio frequency (RF) device, a sensor device, a micro-electro-mechanical-system (MEMS) device, a signal processing device (e.g., digital signal processing (DSP) die), a front-end device (e.g., analog front-end (AFE) dies), combinations thereof (e.g., a system-on-a-chip (SoC) die), or the like.

[0016] In some embodiments, the first semiconductor die **110** includes a first semiconductor substrate **111**, first devices **112** formed in/on the first semiconductor substrate **111**, a first interconnect structure **113** formed over the first semiconductor substrate **111** and electrically coupled to the first devices **112**, and a first bonding structure **114** formed over and electrically coupled to the first interconnect structure **113**. The first semiconductor substrate **111** may be a substrate of silicon, doped or undoped, or an active layer of a semiconductor-on-insulator (SOI) substrate. The first semiconductor substrate **111** may include other semiconductor materials, such as germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon-germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; or combinations thereof. Other suitable substrate, such as a multi-layered substrate or a gradient substrate, may be used.

[0017] The first semiconductor substrate **111** may include a front side **111a** and a back side **111b** opposite to the front side **111a**. For example, the first devices **112** are formed at the front side **111a** of the first semiconductor substrate **111**. The first devices **112** may include active devices (e.g., transistors, diodes, etc.), passive devices (e.g., capacitors, resistors, inductors, etc.), a combination thereof, or the like. Although a single first device **112** is schematically illustrated in the first semiconductor die **110**, it should be noted that the number and the type of the first device **112** may have a different number and type than shown.

[0018] With continued reference to FIG. 1A, the first interconnect structure **113** may be formed over the front side **111a** of the first semiconductor substrate **111** and electrically coupled to the first devices **112** to form integrated circuits. The first interconnect structure **113** may include one or more first dielectric layer(s) **1131** and first metallization patterns **1132** embedded in the first dielectric layer **1131**. The material of the first dielectric layer **1131** may include an oxide (e.g., silicon oxide or aluminum oxide), a nitride (e.g., silicon nitride), a carbide (e.g., silicon carbide), the like, or combinations thereof. The respective first metallization pattern **1132** may include conductive pads, conductive lines, conductive vias, combinations thereof, and/or the like. The respective first metallization pattern **1132** may be formed of a conductive material such as copper, cobalt, aluminum, gold, combinations thereof, or the like. It should be noted

that the first dielectric layer **1131** and the first metallization patterns **1132** may have a different configuration than shown.

[0019] In some embodiments, the first bonding structure **114** includes one or more first bonding dielectric layer(s) **1141** and first bonding connectors **1142** embedded in the first bonding dielectric layer **1141**. The first bonding dielectric layer **1141** may be formed of a material suitable for subsequent dielectric-to-dielectric bonding, such as silicon oxide, silicon oxynitride, and/or the like. The first bonding connectors **1142** may be formed of a conductive material such as copper, aluminum, or the like. The respective first bonding connector **1142** may be a conductive pad, a conductive via, a combination thereof, etc. In some embodiments, the first bonding connectors **1142** are electrically connected to the first metallization patterns **1132** of the first interconnect structure **113**. It should be noted that the first bonding dielectric layer **1141** and the first bonding connectors **1142** may have a different configuration/distribution than shown. In some embodiments, a planarization process (e.g., a chemical mechanical polish (CMP) process, a grinding process, an etching process, a combination thereof, or the like) is performed such that top surfaces (**1141t** and **1142t**) of the first bonding dielectric layer **1141** and the first bonding connectors **1142** are substantially leveled (or coplanar), within process variations.

[0020] With continued reference to FIG. 1A, the first semiconductor die **110** may include a functional (or active) region **110A**, a seal ring region **110S** surrounding the functional region **110A**, and a peripheral region **110P** surrounding the seal ring region **110S**. For example, the seal ring region **110S** is between the functional region **110A** and the peripheral region **110P**. In some embodiments, the peripheral region **110P** is viewed as a scribe line region. In some embodiments, the first devices **112**, the first metallization patterns **1132**, and the first bonding connectors **1142** are located within the functional region **110A**. In some embodiments, both of the first bonding dielectric layer **1141** and the first dielectric layer **1131** extend across the functional region **110A** and the seal ring region **110S**, as well as the peripheral region **110P**.

[0021] In some embodiments, one or more seal ring(s) **115** may be embedded in the first dielectric layer **1131** and within the seal ring region **110S**. In some embodiments, the respective seal ring **115** is disposed in a loop encircling first metallization patterns **1132** in the functional region **110A**. The seal rings **115** may include conductive vias and conductive pads vertically stacked and connected together by the conductive vias, where the conductive pads of the seal rings **115** may be at a same level as the conductive pads of the first metallization patterns **1132**, and the conductive vias of the seal rings **115** may be at the same level as the conductive vias of the first metallization patterns **1132**. It should be noted that the seal rings **115** may have a different configuration than shown.

[0022] In some embodiments, the first bonding structure **114** includes additional bonding connectors **1142D** embedded in the first bonding dielectric layer **1141** and disposed over the seal rings **115** within the seal ring region **110S**. The additional bonding connectors **1142D** may be formed at the same level as the first bonding connectors **1142**. In some embodiments, the additional bonding connectors **1142D** are electrically and spatially isolated from the seal rings **115** at least through the first bonding dielectric layer **1141**. Alter-

natively, the additional bonding connectors **1142D** are physically connected to the underlying seal rings **115**. In some embodiments, the additional bonding connectors **1142D** are dummy connectors and electrically floating in the first semiconductor die **110**. For example, the presence of the additional bonding connectors **1142D** helps to increase the pattern uniformity and metal density, thereby facilitating the subsequent bonding process. Alternatively, the additional bonding connectors **1142D** are omitted, and no conductive feature is formed over the seal ring **115** within the seal ring region **110S**.

[0023] Still referring to FIG. 1A, additional seal rings **1151** may be embedded in the first dielectric layer **1131** and within the peripheral region **110P**. The additional seal rings **1151** may be formed at the same level as the seal rings **115**. It should be noted that the additional seal rings **1151** may have a different configuration than shown. In some embodiments, the additional bonding connectors **1142D** are distributed within the peripheral region **110P** and over the additional seal rings **1151**. The additional bonding connectors **1142D** may be electrically and spatially isolated from the additional seal rings **1151** at least through the first bonding dielectric layer **1141**. The additional bonding connectors **1142D** may (or may not) be physically connected to the additional seal rings **1151**. Alternatively, the additional seal rings **1151** disposed within the peripheral region **110P** and/or the additional bonding connectors **1142D** disposed within the peripheral region **110P** are omitted. The first semiconductor die **110** may (or may not) include any metallization patterns and/or conductive features outside of the seal rings **115** (e.g., within the peripheral region **110P**).

[0024] FIG. 1B illustrates a schematic cross-sectional view of a semiconductor wafer, in accordance with some embodiments. It should be noted that FIG. 1B is provided for illustrative purposes only, and the semiconductor wafer may utilize fewer or additional elements according to some embodiments. Referring to FIG. 1B, a semiconductor wafer **1200** may be provided. The semiconductor wafer **1200** may include a second semiconductor substrate **121** having a front side **121a** and a back side **121b**, a second interconnect structure **123** formed over the front side **121a** of the second semiconductor substrate **121**, a second bonding structure **124** formed over the second interconnect structure **123**, and through vias **125** formed in the second semiconductor substrate **121** and extending into the second interconnect structure **123**.

[0025] The second semiconductor substrate **121** may be a bulk semiconductor substrate, a SOI substrate, a multi-layered semiconductor substrate, or the like. The material of the second semiconductor substrate **121** may be selected from the same group of candidate materials for forming the first semiconductor substrate **111** discussed in FIG. 1A. The second semiconductor substrate **121** may be doped or undoped. In some embodiments, the semiconductor wafer **1200** is free of active/passive devices, and the second semiconductor substrate **121** does not include active/passive devices formed at the front side **121a**. In some embodiments, second devices (e.g., transistors, diodes, capacitors, resistors, inductors, combinations thereof, and/or the like; not shown) are formed at the front side **121a** of the second semiconductor substrate **121**. The second interconnect structure **123** may include one or more second dielectric layer(s) **1231** and second metallization patterns **1232** embedded in the second dielectric layer **1231**. The second dielectric layer

1231 and the second metallization patterns **1232** may be respectively similar to the first dielectric layer **1131** and the first metallization patterns **1132** which are described in FIG. 1A, and thus the details thereof are not repeated herein.

[0026] The second bonding structure **124** may be formed over and electrically connected to the second interconnect structure **123**. For example, the second bonding structure **124** includes one or more second bonding dielectric layer(s) **1241** and second bonding connectors **1242** embedded in the second bonding dielectric layer **1241**. The second bonding connectors **1242** may be electrically connected to the second metallization patterns **1232**. The second bonding dielectric layer **1241** and the second bonding connectors **1242** may be respectively similar to the first bonding dielectric layer **1141** and the first bonding connectors **1142** which are described in FIG. 1A, and thus the details thereof are not repeated herein. In some embodiments, the second bonding structure **124** includes additional bonding connectors **1242D** embedded in the second bonding dielectric layer **1241**. The additional bonding connectors **1242D** may be formed at the same level as the second bonding connectors **1242**. In some embodiments, the additional bonding connectors **1242D** are dummy connectors and electrically isolated from the second bonding connectors **1242**. The additional bonding connectors **1242D** may be electrically floating in the semiconductor wafer **1200**. In some embodiments, the additional bonding connectors **1242D** are subsequently bonded to the additional bonding connectors **1142D** of the first semiconductor die **110**. A planarization process (e.g., a CMP process, a grinding process, an etching process, a combination thereof, or the like) is optionally performed on the second bonding structure **124**, such that top surfaces (**1241t** and **1242t**) of the second bonding dielectric layer **1241**, the second bonding connectors **1242**, and the additional bonding connectors **1142D** are substantially leveled (or coplanar), within process variations.

[0027] The through vias **125** may be formed in the second semiconductor substrate **121** by depositing one or more diffusion barrier layer(s) or isolation layer(s), depositing a seed layer, and depositing a conductive material (e.g., tungsten, titanium, aluminum, copper, any combinations thereof and/or the like) into the trenches of the second semiconductor substrate **121**. For example, the respective through via **125** includes a first end **125a** physically and electrically connected to one of the second metallization patterns **1232** and a second end **125b** opposite to the first end **125a**, where the second end **125b** may be buried in the second semiconductor substrate **121** at this stage.

[0028] FIGS. 2A-2F illustrate schematic cross-sectional views of intermediate steps during a process for forming an integrated circuit (IC) package including a semiconductor structure, in accordance with some embodiments. Unless specified otherwise, the first semiconductor die **110** and the semiconductor wafer **1200** in these embodiments are essentially the same as the like components, which are denoted by like reference numerals in the embodiments shown in FIGS. 1A-1B. The details regarding the first semiconductor die **110** and the semiconductor wafer **1200** may be found in the discussion of the previous embodiments.

[0029] Referring to FIG. 2A and with reference to FIGS. 1A-1B, the first semiconductor die **110** may be bonded to the semiconductor wafer **1200**. Although a single first semiconductor die **110** is illustrated, any number of the first semiconductor die **110** may be bonded to the semiconductor

wafer **1200**. In some embodiments, the first semiconductor die **110** and the semiconductor wafer **1200** are directly bonded in a face-to-face manner by dielectric-to-dielectric bonding and metal-to-metal bonding. For example, the front side **110F** of the first semiconductor die **110** is bonded to the front side **1200F** of the semiconductor wafer **1200**. In some embodiments, the first bonding dielectric layer **1141** is fused to the second bonding dielectric layer **1241** through dielectric-to-dielectric bonding, and dielectric-to-dielectric (e.g., oxide-to-oxide) bonds may be formed therebetween. The first bonding connectors **1142** may be directly bonded to the second bonding connectors **1242** through metal-to-metal bonding, and metal-to-metal (e.g., copper-to-copper) bonds may be formed therebetween. In some embodiments, dielectric-to-metal (e.g., oxide-to-copper; not individually shown) bonds are formed at the bonding interface **IF10** of the first semiconductor die **110** and the semiconductor wafer **1200**. In some embodiments, the bonding interface **IF10** is free of solder material. The bonding interface **IF10** may be substantially flat and planar, within process variations.

[0030] In some embodiments, the bonding of the first semiconductor die **110** and the semiconductor wafer **1200** includes a pre-bonding process and an annealing process. During the pre-bonding process, a force may be applied to press the first semiconductor die **110** against the semiconductor wafer **1200**. The bonding strength of the first and second bonding dielectric layers (**1141** and **1241**) may be improved in the annealing process, in which the first and second bonding dielectric layers (**1141** and **1241**) are annealed at a high temperature. In some embodiments, after the bonding process, the first and second bonding connectors (**1142** and **1242**) are directly connected to one another with a one-to-one correspondence. In some embodiments, the additional bonding connectors (**1142D** and **1242D**) are directly bonded to one another with a one-to-one correspondence.

[0031] It is appreciated that a problem that affects the electrical reliability of the bonded structure is the adhesion between the first semiconductor die **110** and the semiconductor wafer **1200**. Poor adhesion may lead to delamination. In some cases, during the bonding process, the first and second bonding connectors may be expanded under the annealed temperature and apply stresses to the surrounding the first and second bonding dielectric layers, and hence cause delamination. For example, after the bonding process, non-bond areas **NB1** exist at the bonding interface **IF10** (e.g., corresponding to the peripheral region **110P**). During subsequent processing steps (e.g., the formation of an insulating encapsulant described in FIG. 2C), the large CTE mismatch between the insulating encapsulant and the semiconductor die/wafer may generate stress in the resulting structure, especially at the interface between the insulating encapsulant and the semiconductor die/wafer. Under the thermal mismatch stresses, the non-bond areas **NB1** may be enlarged and cracks (if exist) may extend toward the functional region **110A**. This may cause the first semiconductor die and the semiconductor wafer to separate and render the resulting structure to be non-functional or failure. Thus, in the manufacture of the semiconductor structure, it is important to prevent the bonding interface from delaminating and prevent any cracks extending into the functional region **110A**. As described in greater detail below, by partially removing the bonded structure, the bonding interface stress

during the formation of the insulating encapsulant may be reduced, and the adhesion of the bonded structure may be improved.

[0032] Referring to FIG. 2B and with reference to FIG. 2A, a portion of the first semiconductor die **110** in the peripheral region **110P** may be removed by any suitable method to form a first semiconductor die **110'** including a ledge portion **110G**. For example, a photoresist (not shown) is formed on the bonded structure and covers the back side **111b** of the first semiconductor substrate **111** by spin-coating, spray-coating, or any suitable deposition process, and then a patterning process is performed on the photoresist to form an opening by lithography, or the like, where the opening of the photoresist may accessibly expose a portion of the first semiconductor substrate **111** to be removed. Next, the portion of the first semiconductor substrate **111** exposed by the opening of the photoresist may be removed by, for example, plasma etching, laser grooving, and/or any suitable removal process. In some embodiments, not only the portion of the first semiconductor substrate **111** in the peripheral region **110P** which is directly above the non-bond areas **NB1** (if exist), but also a portion of the first dielectric layer **1311** underlying the portion of the first semiconductor substrate **111** may be removed. Afterwards, the photoresist may be removed.

[0033] As shown in FIG. 2B, the first semiconductor die **110'** may include a first portion **110X** and a second portion **110Y** connected to the first portion **110X** and bonded to the semiconductor wafer **1200**. The first portion **110X** may be the rest of the first semiconductor substrate **111'**, and the second portion **110Y** may include the first interconnect structure **113** and the underlying first bonding structure **114**. The first interconnect structure **113** and the underlying first bonding structure **114** may be laterally protruded from the first semiconductor substrate **111'**. The portions of the first interconnect structure **113** and the underlying first bonding structure **114** protruded from the first semiconductor substrate **111'** may be viewed as the ledge portion **110G**. The additional seal ring **1151** and the additional bonding connectors **1142D** may be disposed in the ledge portion **110G**. For example, the first portion **110X** has a lateral dimension **LX1** less than a lateral dimension **LY1** of the second portion **110Y**. The difference of the lateral dimensions (**LY1** and **LX1**) may be the lateral dimension **LG1** of the ledge portion **110G**. It should be noted that the lateral dimensions (**LX1**, **LY1**, and **LG1**) may vary depending on the process and product requirements and construe no limitation in the disclosure. In some embodiments, the sidewall **111W** of the first semiconductor substrate **111'** is laterally displaced from the sidewall **1131W** of the first dielectric layer **1311** and the sidewall **1141W** of the first bonding dielectric layer **1141**, where the sidewall **1141W** is substantially aligned (or coplanar) with the sidewall **1131W**, within process variations. In some embodiments, an upper surface **1131U** of the first dielectric layer **1311** connected to the sidewall **1131W** is accessibly revealed at this stage.

[0034] Referring to FIG. 2C and with reference to FIG. 2B, an insulating encapsulant **132** may be formed on the semiconductor wafer **1200** to cover the first semiconductor die **110'**. In some embodiments, the insulating encapsulant **132** is formed of a molding material or compound and may be formed by compression molding, transfer molding, or the like. The molding material includes a polymer material and optionally includes fillers (not individually illustrated),

where the fillers may be particles of silica or the like, and the polymer material may be an epoxy or the like. The fillers mixed in the polymer material may provide mechanical strength and thermal dispersion for the insulating encapsulant 132. For example, the insulating material is formed over the top surface 1241*t* of the second bonding dielectric layer 1241 of the semiconductor wafer 1200, and the first semiconductor die 110' may be buried or covered by the insulating material. The insulating material may then be cured to form the insulating encapsulant 132.

[0035] A planarization process (e.g., CMP, grinding, etching, combinations thereof, or the like) is optionally performed to planarize the insulating encapsulant 132. The planarization process may (or may not) remove the insulating encapsulant 132 on the back side 111*b* of the first semiconductor substrate 111'. In some embodiments, the back side 111*b* of the first semiconductor die 110' is accessibly exposed by the planarization of the insulating encapsulant 132, and surfaces (e.g., 111*b* and 132*t*) of the first semiconductor die 110' and the insulating encapsulant 132 are substantially leveled (or coplanar), within process variations. In some embodiments, the insulating encapsulant 132 laterally covers the first and second portions (110*X* and 110*Y*) of the first semiconductor die 110'. The insulating encapsulant 132 may be in physical contact with the sidewall 111*W* of the first semiconductor substrate 111', the upper surface 1131*U* and the sidewall 1131*W* of the first dielectric layer 1131, and the sidewall 1141*W* of the first bonding dielectric layer 1141. By partially removing the first semiconductor die 110 to form the first semiconductor die 110' having the ledge portion 110*G*, bonding interface stress, especially in the peripheral region 110*P* of the first semiconductor die 110, may be reduced during the formation of the insulating encapsulant 132. In this manner, even if the non-bond areas (e.g., NB1 labeled in FIG. 2A) and/or cracks exist in the bonded structure, stress in the bonded structure may be relaxed during the formation of the insulating encapsulant 132, thereby preventing delamination/cracks from occurring, becoming more severe, and/or extending into the functional region 110A.

[0036] With continued reference to FIG. 2C and FIG. 2B, a thinning process (e.g., grinding, CMP, etching, combinations thereof, or the like) may be performed on the back side of the semiconductor wafer 1200. For example, the back side 121*b* of the second semiconductor substrate 121 is thinned down until at least a portion of the second ends 125*b* of the through vias 125 is accessibly exposed. In some embodiments, the thinning process is performed after the formation of the insulating encapsulant 132. Since the through vias 125 penetrate through the second semiconductor substrate 121, the through vias 125 may be viewed as through-substrate vias (TSVs) 125.

[0037] Referring to FIG. 2D and with reference to FIG. 2C, a plurality of conductive terminals 142 may be formed over the back side 121*b* of the second semiconductor substrate 121 and electrically connected to the TSVs 125. The conductive terminals 142 may be controlled collapse chip connection (C4) bumps, ball grid array (BGA) connectors, solder balls, metal pillars, micro-bumps, electroless nickel-electroless palladium-immersion gold (ENEPIG) formed bumps, or the like. The conductive terminals 142 may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the conductive

terminals 142 are formed by forming a solder material; and performing a reflow process on the solder material to form desired bump shapes. In some embodiment, the respective conductive terminal 142 includes a pillar portion (e.g., a copper pillar) and a cap portion formed on the pillar portion, where the pillar portion has a substantially vertical sidewall and the cap portion has a bump profile.

[0038] In some embodiments, before forming the conductive terminals 142, a redistribution structure 150 is formed on the back side 121*b* of the second semiconductor substrate 121 and the second ends 125*b* of the TSVs 125. For example, the redistribution structure 150 includes one or more dielectric layer(s) 151 and conductive patterns (or redistribution wirings) 152 formed in the dielectric layer 151 and electrically connected to the TSVs 125. The dielectric layer 151 may be formed of any suitable dielectric material such as polybenzoxazole (PBO), polyimide, a benzocyclobuten (BCB), a combination thereof, or the like. The conductive patterns 152 may include conductive pads, conductive vias, conductive lines, a combination thereof, or the like, and may be formed of any suitable conductive material such as copper, cobalt, aluminum, gold, combinations thereof, or the like. In some embodiments, the conductive patterns 152 include under bump metallization (UBM) pads, and the conductive terminals 142 may be formed on the UBM pads. Alternatively, the redistribution wirings of the redistribution structure 150 is omitted. In such cases, the UBM pads are directly formed on the second ends 125*b* of the TSVs 125, and the conductive terminals 142 are formed on the UBM pads to be electrically coupled to the TSVs 125.

[0039] Referring to FIG. 2E and with reference to FIG. 2D, a singulation process is optionally performed by cutting along scribe lanes SL1 to form individual semiconductor structure 10A. For example, the semiconductor structure 10A includes a first tier 101 stacked upon a second tier 102, where the first tier 101 includes the first semiconductor die 110' and the insulating encapsulant 132 covering the first semiconductor die 110', and the second tier 102 includes the second semiconductor die 120 formed by singulating the semiconductor wafer 1200, the redistribution structure 150 underlying the second semiconductor die 120, and the conductive terminals 142 electrically coupled to the second semiconductor die 120 through the redistribution structure 150. In some embodiments, the second semiconductor die 120 is viewed as an interposer. In some embodiments, the scribe lanes SL1 vertically pass through the periphery (e.g., the ledge portion 110*G* or the second portion 110*Y*) of the first semiconductor die 110'. For example, at least the edge of the ledge portion 110*G* is removed after the singulation process to form the ledge portion 110*G*'. The portion of the insulating encapsulant 132 laterally surrounding the second portion 110*Y* of the first semiconductor die 110' may be removed during the singulation process, and the rest of the insulating encapsulant 132 laterally surrounding the first portion 110*X* of the first semiconductor die 110' may remain after the singulation process. After the singulation process, the maximum lateral dimension LX1 of the first semiconductor substrate 111' of the first semiconductor die 110' is less than the maximum lateral dimension LM2 the second semiconductor die 120. The maximum lateral dimension of the first interconnect structure 113 may be substantially equal to the maximum lateral dimension LM2 the second semiconductor die 120.

[0040] As shown in FIG. 2E, the singulated insulating encapsulant 132' may have a singulated sidewall 132W substantially aligned (or coplanar) with the singulated sidewall 110YW of the second portion 110Y of the first semiconductor die 110' and the singulated sidewall 120W of the second semiconductor die 120, within process variations. For example, the singulated sidewall 110YW of the second portion 110Y of the first semiconductor die 110' includes the singulated sidewall 113W' of the first interconnect structure 113 and the singulated sidewall 114W' of the first bonding structure 114. The locations of the scribe lanes SL1 may be adjusted depending on process and product requirements. For example, the scribe lanes SL1 only pass through the insulating encapsulant 132 and the underlying semiconductor wafer 1200 without passing through the first semiconductor die 110'. In such cases, the first and second portions (110X and 110Y) of the first semiconductor die 110' remain laterally covered by the singulated insulating encapsulant. In alternative embodiments, the scribe lanes SL1 pass through the ledge portion 110G, and the additional seal rings 1151 in the ledge portion 110G are singulated as will be described later in accompanying with FIG. 3B.

[0041] Referring to FIG. 2F and with reference to FIG. 2E, the semiconductor structure 10A is optionally mounted on a package substrate 20 using the conductive terminals 142 to form an integrated circuit (IC) package 30. The package substrate 20 may include a substrate 202, which may be made of a semiconductor material such as silicon, germanium, diamond, or the like. Alternatively, compound materials such as silicon germanium, silicon carbide, gallium arsenic, indium arsenide, indium phosphide, silicon germanium carbide, gallium arsenic phosphide, gallium indium phosphide, combinations of these, and the like, may also be used. The substrate 202 may be a SOI substrate. Alternatively, the substrate 202 includes an insulating core (not individually illustrated) such as a fiberglass reinforced resin core (e.g., FR4), BT resin core or includes printed circuit board (PCB) materials or films. Build up films (e.g., Ajinomoto build-up film or other laminates; not individually illustrated) may be used for the substrate 202. The substrate 202 may include active and/or passive devices (not illustrated) to generate the functional requirements of the design for the system.

[0042] The package substrate 20 may include contact pads 204 formed in/on the substrate 202. The conductive terminals 142 may be reflowed to attach the semiconductor structure 10A to the contact pads 204 of the package substrate 20. After coupling the conductive terminals 142 to the contact pads 204, the semiconductor structure 10A may be electrically coupled to the package substrate 20. In some embodiments, the IC package 30 includes an underfill 206 formed in a gap between the semiconductor structure 10A and the package substrate 20. The underfill 206 may laterally surround the conductive terminals 142 for protection. The underfill 206 may be formed by a capillary flow process after the semiconductor structure 10A is attached or may be formed by a suitable deposition method before the semiconductor structure 10A is attached. The underfill 206 may be a continuous material extending from the gap between the package substrate 20 and the semiconductor structure 10A. In some embodiments, the underfill 206 extends upward to be in physical contact with the singulated sidewall 120W of the second semiconductor die 120. Depending on the applied amount of the underfill 206, in some embodiments,

the underfill 206 extends upward to be in physically contact with the singulated sidewall 110YW of the first semiconductor die 110'. The perimeter of the bonding interface IF10 may be surrounded by the underfill 206. In some other embodiments, the underfill 206 extends upward to be in physically contact with the singulated sidewall 132W of the singulated insulating encapsulant 132'. The above examples are provided for illustrative purposes only, and the IC package 30 may include fewer or additional elements, in other embodiments.

[0043] FIG. 3A and FIG. 3B illustrate schematic cross-sectional views of variations of a semiconductor structure, in accordance with some embodiments. Unless specified otherwise, the materials and the formation methods of the components in these embodiments are essentially the same as the like components, which are denoted by like reference numerals in the embodiments shown in FIG. 2E. The details of the components shown in FIGS. 3A-3B may be found in the discussion of the previous embodiments.

[0044] Referring to FIG. 3A and with reference to FIG. 2E, a semiconductor structure 10B shown in FIG. 3A is similar to the semiconductor structure 10A shown in FIG. 2E, except that the first semiconductor die 110'-1 of the first tier 101' includes a third portion 110Z vertically interposed between the first portion 110X and the second portion 110Y', and the additional seal ring 1151 in the second portion 110Y' is in direct contact with the singulated insulating encapsulant 132'-1. The third portion 110Z may have a lateral dimension substantially equal to the lateral dimension of the first portion 110X and less than the lateral dimension of the second portion 110Y'. For example, the third portion 110Z is a part of the first interconnect structure 113', and the second portion 110Y' includes the rest part of the first interconnect structure 113' and the underlying first bonding structure 114. The second portion 110Y' may be laterally protruded from the third portion 110Z, and the protruded part may be viewed as the ledge portion 110G'-1. The additional seal ring 1151 may be disposed in the second portion 110Y' and in the ledge portion 110G'-1.

[0045] In some embodiments, during the process of partially removing the first semiconductor die 110 as described in FIG. 2B, a portion of the first interconnect structure 113 directly underlying the portion of the first semiconductor substrate 111 in the peripheral region 110P (labeled in FIG. 2A) is also removed until at least a portion of the additional seal ring 1151 is accessibly exposed by the first dielectric layer 1131'. The additional seal ring 1151 may act as a stop layer during the removal process. In some embodiments, the upper surface 1131U' of the first dielectric layer 1131' and the upper surface 1151U' of the additional seal ring 1151 are substantially leveled (or coplanar), within process variations. In some embodiments, the upper surface 1151U' of the additional seal ring 1151 is slightly protruded from the upper surface 1131U' of the first dielectric layer 1131'. The sidewall 1131W' of the first dielectric layer 1131' connected to the upper surface 1131U' may be substantially leveled (or coplanar) with the sidewall 111W of the first semiconductor substrate 111'. The sidewall 1131W' of the first dielectric layer 1131' may be viewed as a sidewall of the third portion 110Z. The singulated sidewall 110YW of the second portion 110Y' may be laterally displaced from the sidewall 1131W' of the third portion 110Z.

[0046] As shown in FIG. 3A, the first tier 101' of the semiconductor structure 10B may include the singulated

insulating encapsulant **132'-1** extending along the sidewalls (**111W** and **1131W'**) of the first portion **110X** and the third portion **110Z**. The singulated insulating encapsulant **132'-1** may be in physical contact with the upper surface **1131U'** of the first dielectric layer **1131'** and the upper surface **1151U'** of the additional seal ring **1151**. After the singulation process, the singulated sidewall **132W** of the singulated insulating encapsulant **132'-1** may be substantially aligned (or coplanar) with the singulated sidewall **110YW** of the second portion **110Y'** and the singulated sidewall **120W** of the second semiconductor die **120**. The semiconductor structure **10B** is optionally mounted on the package substrate **20** (see FIG. 2F) using the conductive terminals **142** to form an IC package.

[0047] Referring to FIG. 3B and with reference to FIG. 2E, a semiconductor structure **10C** shown in FIG. 3B is similar to the semiconductor structure **10A** shown in FIG. 2E, except that the second portion **110Y** of the first semiconductor die **110'-2** may have a singulated sidewall **110YW'** including the singulated sidewall **1151W** of the additional seal ring **1151'**. For example, the scribe lanes **SL1** (labeled in FIG. 2D) pass through the additional seal ring **1151** in the ledge portion **110G** of the first semiconductor die **110'**. In such cases, during the singulation process, the cutting tool (e.g., a blade or the like) may cut through insulating encapsulant **132**, the first dielectric layer **1131**, the additional seal ring **1151**, and the first bonding structure **114** so as to form the first tier **101'**. In some embodiments, the scribe lanes **SL1** also pass through the additional bonding connectors (**1142D** and **1242D**) directly below the singulated additional seal ring **1151** in the ledge portion **110G**. In such cases, the singulated sidewalls of the additional bonding connectors (**1142D** and **1242D**) may be exposed at an outer sidewall of the semiconductor structure **10C**. Alternatively, no additional bonding connectors (**1142D** and **1242D**) are singulated. Thus, the singulated bonding connectors (**1142D** and **1242D**) in FIG. 3B are illustrated in the dashed lines to indicate they may (or may not) exist.

[0048] After the singulation process, the singulated sidewall **110YW'** of the second portion **110Y** of the first semiconductor die **110'-2** may include the singulated sidewall **1131W''** of the first dielectric layer **1131**, the singulated sidewall **1151W** of the additional seal ring **1151'**, and the singulated sidewall **114W'** of the first bonding structure **114** that are substantially aligned (or coplanar) with one another, within process variations. The singulated sidewall **110YW'** of the second portion **110Y** of the first semiconductor die **110'-2** may be substantially aligned (or coplanar) with the singulated sidewall **132W** of the singulated insulating encapsulant **132'** and the singulated sidewall **120W** of the second semiconductor die **120**, within process variations. The semiconductor structure **10C** is optionally mounted on the package substrate **20** (see FIG. 2F) using the conductive terminals **142** to form an IC package.

[0049] It is appreciated that depending on the locations of the scribe lanes **SL1** (labeled in FIG. 2D), the singulated sidewall of the semiconductor structure may have a different configuration than shown. For example, the lateral dimension **LG1'** of the ledge portion **110G'-2** is non-zero. In some embodiments, the lateral dimension **LG1'** is greater than 1 μm . Other values are also possible. In some embodiments, the scribe lanes **SL1** (labeled in FIG. 2D) pass through both of conductive pads and conductive vias of the respective additional seal ring **1151**. As illustrated by FIG. 3B, the

singulated sidewall **1151W** of the respective additional seal ring **1151'** may thus include the sidewalls of the conductive pads and the sidewalls of the conductive vias that are substantially aligned (or coplanar) with one another, within process variations. In some other embodiments, the scribe lanes **SL1** (labeled in FIG. 2D) pass through the conductive pads of the respective additional seal ring **1151** without passing through the conductive vias of the respective additional seal ring **1151**. In such cases, the singulated sidewall **1151W** of the respective additional seal ring **1151'** may include the sidewalls of the conductive pads, and the sidewalls of the conductive pads of the additional seal ring **1151'** and the segmental sidewalls of the first dielectric layer **1131** are vertically and alternately arranged. In alternative embodiments, the scribe lanes **SL1** (labeled in FIG. 2D) pass through the first dielectric layer **1311**, and the additional seal rings **1151** are outside the scribe lanes **SL1**. In such cases, during the singulation process, the additional seal rings **1151** are completely removed. Accordingly, the additional seal rings **1151'** are encircled by the dashed boxes outlined in FIG. 3B to indicate that those may (or may not) exist in the ledge portion of the first semiconductor die **110'-2**.

[0050] FIGS. 4A-4C illustrate schematic cross-sectional views of intermediate steps during a process for forming a semiconductor structure, in accordance with some embodiments. Unless specified otherwise, the materials and the formation methods of the components in these embodiments are essentially the same as the like components, which are denoted by like reference numerals in the embodiments shown in FIGS. 2A through 2E. The details regarding the formation process and the materials of the components shown in FIGS. 4A-4C may be found in the discussion of the previous embodiments.

[0051] Referring to FIG. 4A and with reference to FIGS. 2A-2B, the first semiconductor die **110** may be bonded to the semiconductor wafer **1200** as described in FIG. 2A. After the bonding process, a portion of the bonded structure may be removed by any suitable methods (e.g., plasma etching, laser grooving, combinations thereof, other patterning process, or the like). For example, the peripheral region **110P** of the first semiconductor die **110** is partially (or entirely) removed to form a first semiconductor die **110''** having a continuous sidewall **110W**. The continuous sidewall **110W** may include the sidewall **111V** of the first semiconductor substrate **111**, the sidewall **1131V** of the first dielectric layer **1131**, and the sidewall **1141V** of the first bonding structure **114**. For example, the sidewalls (**111V**, **1131V**, and **1141V**) are substantially aligned (or coplanar) with one another, within process variations.

[0052] In some embodiments, a portion of the first semiconductor die **110** corresponding to the non-bond areas **NB1** (if exist; labeled in FIG. 2A) is removed. In some embodiments, the first semiconductor substrate **111**, the first interconnect structure **113** underlying the first semiconductor substrate **111**, and the first bonding structure **114** underlying the first interconnect structure **113** that are within the peripheral region **110P** are partially (or fully) removed. In some embodiments, the additional seal rings **1151** disposed in the peripheral region **110P** and the additional bonding connectors **1142D** (if exist) directly underneath the additional seal rings **1151** are also removed. Alternatively, the additional seal rings **1151** and/or the additional bonding connectors **1142D** within the peripheral region **110P** may be partially removed. In such cases, a sidewall of the additional seal

rings **1151** and/or a sidewall of the additional bonding connectors **1142D** may be exposed at an outer sidewall of the first semiconductor die **110"**. For example, the peripheral region **110P** of the first semiconductor die **110** is partially removed, and the lateral dimension **110PL** of the remaining peripheral region **110P'** is non-zero. For example, the lateral dimension **110PL** of the remaining peripheral region **110P'** is greater than 1 μm . Other values are also possible.

[0053] With continued reference to FIG. 4A, a portion of the semiconductor wafer **1200** directly underlying the portion of the first semiconductor die **110** in the peripheral region **110P** may also be removed to form a semiconductor wafer **1200'** having a recess **120R**. The recess **120R** may be a close loop surrounding the first semiconductor die **110"** in a top view (not shown). The depth of the recess **120R** may vary depending on the process and product requirements as long as the non-bond areas **NB1** (labeled in FIG. 2A) of the bonded structure are removed. The width of the recess **120R** may correspond to the width of the removed portion of the first semiconductor die **110**. By removing the non-bond areas **NB1** (labeled in FIG. 2A) in the first semiconductor die and the semiconductor wafer, the rest of the bonding interface **IF10'** may remain well-bonded. For example, the bottommost part of the recess **120R** reaches the interface of the second bonding structure **124'** and the second interconnect structure **123'**. In the illustrated embodiment, the recess **120R** is defined by the inner sidewalls (**124V1** and **124V2**) of the second bonding dielectric layer **1241'** of the second bonding structure **124'**, the inner sidewalls (**123V1** and **123V2**) of the second dielectric layer **1231'** of the second interconnect structure **123'**, and the upper surface **1231i'** of the second dielectric layer **1231'**. In other embodiments where the bottom of the recess **120R** reaches the second bonding structure without extending into the second interconnect structure, the recess **120R** is defined by the inner sidewalls (**124V1**, **124V2**, **123V1**, and **123V2**) and the upper surface of the second bonding dielectric layer. In some embodiments, the inner sidewalls (**124V1** and **123V1**) are substantially aligned (or coplanar) with the continuous sidewall **110W** of the first semiconductor die **110"**.

[0054] Referring to FIG. 4B and with reference to FIG. 4A and FIGS. 2C-2D, an insulating encapsulant **232** may be formed on the semiconductor wafer **1200'** to cover the first semiconductor die **110"**. The material and the forming method of the insulating encapsulant **232** may be similar to the insulating encapsulant **132** described in FIG. 2C, and thus the details thereof are not repeated herein. The insulating encapsulant **232** may extend along the continuous sidewall **110W** of the first semiconductor die **110"** and fill the recess **120R** of the semiconductor wafer **1200'**. For example, the insulating encapsulant **232** is in physical contact with the inner sidewalls (**124V1**, **124V2**, **123V1**, and **123V2**) of the second bonding dielectric layer **1241'** and the second dielectric layer **1231'** and the upper surface **1231i'** of the second dielectric layer **1231'** that define the recess **120R**. A planarization process (e.g., CMP, grinding, etching, combinations thereof, or the like) is optionally performed to planarize the insulating encapsulant **232** and the first semiconductor die **110"**. In some embodiments, the back side **111b** of the first semiconductor die **110"** and top surface **232i'** of the insulating encapsulant **232** are substantially level (or coplanar), within process variations. Since the non-bond areas **NB1** (labeled in FIG. 2A) are removed before forming

the insulating encapsulant **232**, no delamination/crack exists in the remaining bonding interface **IF10'**. In this manner, the likelihood of delamination/crack propagation caused by the stress induced during the formation of the insulating encapsulant **232** may be reduced or eliminated.

[0055] In some embodiments, a thinning process (e.g., grinding, CMP, etching, combinations thereof, or the like) is performed on the back side of the semiconductor wafer **1200'** until at least a portion of the second ends **125b** of the TSVs **125** is accessibly exposed. The thinning process may be similar to the process described in FIG. 2C, and thus the details thereof are not repeated herein. In some embodiments, the redistribution structure **150** and the conductive terminals **142** are sequentially formed on the back side of the semiconductor wafer **1200'** and the second ends **125b** of the TSVs **125**. The details of the redistribution structure **150** and the conductive terminals **142** are described in FIG. 2D and not repeated herein. In alternative embodiments, the redistribution structure **150** is replaced with the UBM pads to directly couple the conductive terminals **142** to the TSVs **125**.

[0056] Referring to FIG. 4C and with reference to FIG. 4B and FIGS. 2E-2F, a singulation process is optionally performed by cutting along the scribe lanes **SL1** to form individual semiconductor structure **10D**. In some embodiments, the scribe lanes **SL1** vertically pass through the recess **120R** of the semiconductor wafer **1200'**, and the cutting tool (e.g., a saw, a blade, or the like) may travel through the insulating encapsulant **232** and the semiconductor wafer **1200'**. For example, the resulting semiconductor structure **10D** includes a first tier **201** stacked upon a second tier **202**, where the first tier **201** includes the first semiconductor die **110"** and the insulating encapsulant **232'** laterally covering the first semiconductor die **110"**, and the second tier **202** includes the second semiconductor die **120'** formed by singulating the semiconductor wafer **1200'**, the redistribution structure **150** underlying the second semiconductor die **120'**, and the conductive terminals **142** electrically coupled to the second semiconductor die **120'** through the redistribution structure **150**. After the singulation process, the maximum lateral dimension **LX1** of the first semiconductor substrate **111** is less than the maximum lateral dimension **LM2** the second semiconductor die **120**. The maximum lateral dimensions of the first interconnect structure **113**, the first bonding interconnect structure **114**, and the second bonding structure **124** may be substantially equal to the maximum lateral dimension **LX1** of the first semiconductor substrate **111**. The insulating encapsulant **232'** may extend vertically into the second tier **202** and beyond the bonding interface **IF10'**. For example, the insulating encapsulant **232'** extends along the continuous sidewall **110W** of the first semiconductor die **110"** and the inner sidewalls (**124V1** and **123V1**) of the second bonding structure **124'** and the second interconnect structure **123'**.

[0057] In some embodiments, the singulated sidewall **232W** of the insulating encapsulant **232'** is substantially aligned (or coplanar) with the singulated sidewall of the second semiconductor die **120'**. In the embodiments where the scribe lanes **SL1** vertically pass through the recess **120R** of the semiconductor wafer **1200'**, the singulated sidewall of the second semiconductor die **120'** includes the outer sidewall **123W1** of the second interconnect structure **123'**, the outer sidewall **121W** of the second semiconductor substrate **121**, and the outer sidewall **150W** of the redistribution

structure **150** (if exists). The locations of the scribe lanes **SL1** labeled in FIG. 4B may be adjusted depending on process and product requirements. In alternative embodiments, the scribe lanes **SL1** vertically pass through the area outside the recess **120R**, and thus after the singulation process, the recess **120R** filled with the insulating encapsulant **232** may remain in the resulting semiconductor structure. The semiconductor structure **10D** is optionally mounted on the package substrate **20** (see FIG. 2F) using the conductive terminals **142** to form an IC package.

[0058] FIG. 5A illustrates a schematic cross-sectional view of a first semiconductor die, in accordance with some embodiments. It should be noted that FIG. 5A is provided for illustrative purposes only, and the first semiconductor die may utilize fewer or additional elements according to some embodiments. Unless specified otherwise, the first semiconductor die in FIG. 5A is essentially the same as the first semiconductor die described in FIG. 1A. The details of the first semiconductor die shown in FIG. 5A may be found in the discussion of the previous embodiments.

[0059] Referring to FIG. 5A and with reference to FIG. 1A, a first semiconductor die **210** may be similar to the first semiconductor die **110** described in FIG. 1A, except that the first semiconductor die **210** includes a first ledge **210G1** and a second ledge **210G2** which are disposed in the peripheral region **210P**. The first semiconductor die **210** may have a stepped sidewall. For example, the first semiconductor die **210** is formed in a wafer (not shown), which may include different die regions that are singulated in subsequent steps to form a plurality of first semiconductor dies **210**. To perform a singulation process for forming the first semiconductor die **210**, a shallow recess may be formed in the scribe line regions of the semiconductor wafer by etching or other suitable recessing process, thereby forming the sidewall **1141W** of the first bonding dielectric layer **1141**. If the shallow recess is deep enough to reach the first interconnect structure **113"**, the first sidewall **1131W1** of the first dielectric layer **1131"** and the first surface **1131U** of the first dielectric layer **1131"** connected to the first sidewall **1131W1** are also formed. Next, a grooving process (e.g., laser grooving, plasma dicing, or the like) may be performed on the semiconductor wafer and through the shallow recess, thereby forming the groove connected to the recess. For example, the groove penetrates through the first dielectric layer **1311"**, and the second ledge **210G2** having the second sidewall **1131V** of the first dielectric layer **1311"** is formed. In some embodiments, the grooving process stops until the front side **111a** of the first semiconductor substrate **111** is accessibly exposed. Subsequently, a sawing process may be performed on the semiconductor wafer to fully separate the die regions from one another to form individual first semiconductor dies **210**. The sawing process may be performed through the shallow recess and the underlying groove in the scribe line regions, and the first ledge **210G1** having the sidewall **111V** of the first semiconductor substrate **111** is formed.

[0060] The aforementioned steps for forming the first semiconductor die **210** are merely an example, and other suitable methods for forming the first semiconductor die **210** having the stepped sidewall profile may be used. The lateral dimensions (e.g., the widths) of the first and second ledges (**210G1** and **210G2**) may vary and may depend on the width of the groove formed by the grooving process and the blade which is used to perform the sawing process. The lateral

dimensions of the first and second ledges (**210G1** and **210G2**) construe no limitation in the disclosure. Due to differences in the processes of recessing/grooving/sawing, sidewalls/surfaces of different regions of the first semiconductor die **210** may have different roughness. For example, the sidewalls/surfaces (**1131W**, **1131U**, **1131V**, and **111a**) formed by recessing/grooving are smoother than the sidewall **111V** formed by sawing. In some embodiments, a surface roughness of the sidewalls/surfaces (**1131W**, **1131U**, **1131V**, and **111a**) is less than that of the sidewall **111V**.

[0061] FIGS. 5B-5E illustrate schematic cross-sectional views of intermediate steps during a process for forming a semiconductor structure, in accordance with some embodiments. Unless specified otherwise, the materials and the formation methods of the components in these embodiments are essentially the same as the like components, which are denoted by like reference numerals in the embodiments shown in FIGS. 2A through 2E. The details regarding the formation process and the materials of the components shown in FIGS. 5B-5E may be found in the discussion of the previous embodiments.

[0062] Referring to FIG. 5B and with reference to FIG. 5A and FIG. 2A, the first semiconductor die **210** may be bonded to the semiconductor wafer **1200**. The bonding process of the first semiconductor die **210** and the semiconductor wafer **1200** may be similar to the process described in FIG. 2A, and thus the details thereof are not repeated herein. For example, the first bonding structure **114** of the first semiconductor die **210** may be bonded to the second bonding structure **124** of the semiconductor wafer **1200**, and the bonding interface **IF20** of the first semiconductor die **210** and the semiconductor wafer **1200** may be substantially flat and planar.

[0063] Referring to FIG. 5C and with reference to FIG. 5B and FIG. 2B, a portion of the first semiconductor die **210** in the peripheral region **210P** may be removed by any suitable method to form a first semiconductor die **210'**. For example, a photoresist (not shown) is formed on the bonded structure and covers the back side of the first semiconductor substrate **111** by spin-coating, spray-coating, or any suitable deposition process, and then a patterning process may be performed on the photoresist to form an opening by lithography or the like, where the opening of the photoresist may accessibly expose a portion of the first semiconductor substrate **111** to be removed. Next, the portion of the first semiconductor substrate **111** exposed by the opening of the photoresist may be removed by, for example, etching or any suitable removal process. Subsequently, the photoresist may be removed. In some embodiments, only the portion of the first semiconductor substrate **111** in the peripheral region **210P** is removed to accessibly expose the second surface **1131'** of the first dielectric layer **1131"** which is opposite to the first surface **1131U**. In alternative embodiments, not only the portion of the first semiconductor substrate **111** in the peripheral region **210P**, but also a portion of the first dielectric layer **1131"** directly underlying the portion of the first semiconductor substrate **111** in the peripheral region **210P** is removed, as will be described later in accompanying with FIG. 6.

[0064] As shown in FIG. 5C, the first semiconductor die **210'** may include a first portion **210X**, a second portion **210Y** underlying the first portion **210X**, and a third portion **210Z** underlying the second portion **210Y** and bonded to the semiconductor wafer **1200**. For example, the first portion

210X is the remaining first semiconductor substrate **111-1**, the second portion **210Y** is a part of the second interconnect structure **113"**, and the third portion **210Z** includes the rest of the second interconnect structure **113"** and the underlying first bonding structure **114**. In some embodiments, the second portion **210Y** is laterally protruded from the first portion **210X** and the third portion **210Z**, and the third portion **210Z** is wider than the first portion **210X**. For example, the lateral dimension **LY1** of the second portion **210Y** is greater than the lateral dimension **LZ1** of the third portion **210Z**, and the lateral dimension **LZ1** of the third portion **210Z** is greater than the lateral dimension **LX1** of the first portion **210X**. The sidewall **111V'** of the first portion **210X** may be laterally displaced from the sidewall **1131V** of the second portion **210Y**, and the sidewalls (**111V'** and **1131V**) may be laterally displaced from the sidewall (**1131W** and **1141W**) of the third portion **210Z**.

[0065] Referring to FIG. 5D and with reference to FIG. 5C and FIG. 2C-2D, an insulating encapsulant **332** may be formed on the semiconductor wafer **1200** to cover the first semiconductor die **210'**. The material and the forming method of the insulating encapsulant **332** may be similar to the insulating encapsulant **132** described in FIG. 2C, and thus the details thereof are not repeated herein. A planarization process (e.g., CMP, grinding, etching, combinations thereof, or the like) is optionally performed to planarize the insulating encapsulant **332**. In some embodiments, the back side **111b** of the first semiconductor substrate **111-1** and the top surface **332t** of the insulating encapsulant **332** are substantially level (or coplanar), within process variations. The insulating encapsulant **332** may have portions with different widths which correspond to the first/second/third portions of the first semiconductor die **210'**. The insulating encapsulant **332** may extend along the stepped sidewall of the first semiconductor die **210'**. For example, the insulating encapsulant **332** is in physical contact with the sidewall **111V'** of the first semiconductor substrate **111-1**, the first and second surfaces (and **1131t**) and the first and second sidewalls (**1131W** and **1131V**) of the first dielectric layer **1131"**, and the sidewall **1141W** of the first bonding structure **114**.

[0066] In some embodiments, a thinning process (e.g., grinding, CMP, etching, combinations thereof, or the like) is performed on the back side of the semiconductor wafer **1200** until at least a portion of the second ends **125b** of the TSVs **125** is accessibly exposed. The thinning process may be similar to the process described in FIG. 2C, and thus the details thereof are not repeated herein. In some embodiments, the redistribution structure **150** and the conductive terminals **142** are sequentially formed on the back side of the semiconductor wafer **1200** and the second ends **125b** of the TSVs **125**. The details of the redistribution structure **150** and the conductive terminals **142** are described in FIG. 2D and not repeated herein. In alternative embodiments, the redistribution structure **150** is replaced with the UBM pads to directly couple the conductive terminals **142** to the TSVs **125**.

[0067] Referring to FIG. 5E and with reference to FIG. 5D and FIGS. 2E-2F, a singulation process is optionally performed by cutting along the scribe lanes **SL1** to form individual semiconductor structure **10E**. In some embodiments, the scribe lanes **SL1** vertically pass through the periphery of the first semiconductor die **210'**. In such cases, at least a periphery of the second portion **210Y** of the first semiconductor die **210'** may be removed after the singula-

tion process. The portion of the insulating encapsulant **332** laterally surrounding the second portion **210Y** of the first semiconductor die **210'** may be removed during the singulation process, and the rest of the insulating encapsulant **332** laterally surrounding the first portion **210X** and the third portion **210Z** of the first semiconductor die **210'** may remain after the singulation process. The locations of the scribe lanes **SL1** labeled in FIG. 5D may be adjusted depending on process and product requirements. In some other embodiments, the scribe lanes **SL1** only pass through the insulating encapsulant **332** and the underlying semiconductor wafer **1200** without passing through the first semiconductor die **210'**, and thus the first/second/third portions (**210X/210Y/210Z**) of the first semiconductor die **210'** remain covered by the insulating encapsulant **332** after the singulation process. In alternative embodiments, the scribe lanes **SL1** vertically pass through the additional seal rings **1151**, and thus the additional seal rings **1151** are cut during the singulation process, as described and shown in FIG. 3B.

[0068] In some embodiments, the semiconductor structure **10E** includes a first tier **301** stacked upon the second tier **102**, where the first tier **301** includes the first semiconductor die **210'** and the insulating encapsulant **332'** laterally covering the first semiconductor die **210'**, and the second tier **102** includes the second semiconductor die **120** formed by singulating the semiconductor wafer **1200**, the redistribution structure **150** underlying the second semiconductor die **120**, and the conductive terminals **142** electrically coupled to the second semiconductor die **120** through the redistribution structure **150**. After the singulation process, the maximum lateral dimension **LX1** of the first semiconductor substrate **111-1** of the first semiconductor die **210'** is less than the maximum lateral dimension **LM2** the second semiconductor die **120**. The maximum lateral dimension of the first interconnect structure **113"** may be substantially equal to the maximum lateral dimension **LM2** the second semiconductor die **120**. The insulating encapsulant **332'** may include a first portion **3321** laterally covering the first portion **210X** of the first semiconductor die **210'** and a second portion **3322** laterally covering the third portion **210Z** of the first semiconductor die **210'**. The first portion **3321** and the second portion **3322** of the insulating encapsulant **332'** may be vertically separated from each other by the second portion **210Y** of the first semiconductor die **210'**. The singulated sidewalls (**3321V** and **3322V**) of the first and second portions (**3321** and **3322**) of the insulating encapsulant **332'** may be substantially aligned (or coplanar) with the singulated sidewall **1131V'** of the second portion **210Y** of the first semiconductor die **210'** and the singulated sidewall **120W** of the second semiconductor die **120**, within process variations. The semiconductor structure **10E** is optionally mounted on the package substrate **20** (see FIG. 2F) using the conductive terminals **142** to form an IC package.

[0069] FIG. 6 illustrates schematic cross-sectional views of a semiconductor structure, in accordance with some embodiments. Unless specified otherwise, the materials and the formation methods of the components in these embodiments are essentially the same as the like components, which are denoted by like reference numerals in the embodiments shown in FIG. 5E and FIGS. 3A. The details of the components shown in FIG. 6 may be found in the discussion of the previous embodiments.

[0070] Referring to FIG. 6 and with reference to FIG. 5E and FIG. 3A, a semiconductor structure **10F** shown in FIG.

6 is similar to the semiconductor structure 10E shown in FIG. 5E, except that the first semiconductor die 210" of the first tier 301' includes a fourth portion 210W vertically interposed between the first portion 210X and the second portion 210Y', and the additional seal ring 1151 in the second portion 210Y' is in direct contact with the insulating encapsulant 332". The fourth portion 210W of the first semiconductor die 210" may have a lateral dimension substantially equal to the lateral dimension of the first portion 210X and less than the lateral dimension of the second portion 210Y'. For example, the fourth portion 210W is a part of the first interconnect structure 113"-1, and the second portion 210Y' includes the rest part of the first interconnect structure 113"-1. The second portion 210Y' may be laterally protruded from the fourth portion 210W. The additional seal ring 1151 may be disposed in the second portion 210Y'.

[0071] The processes of forming the first semiconductor die 210" may be similar to the processes described in FIG. 3A. For example, during the process of partially removing the first semiconductor die 210 as described in FIG. 5C, a portion of the first interconnect structure 113" underlying the portion of the first semiconductor substrate 111 in the peripheral region 210P (labeled in FIG. 5B) is also removed until at least a portion of the additional seal ring 1151 is accessibly exposed by the first dielectric layer 1131"-1. The additional seal ring 1151 may act as a stop layer during the removal process. In some embodiments, the upper surface 1131U' of the first dielectric layer 1131"-1 and the upper surface 1151U' of the additional seal ring 1151 are substantially leveled (or coplanar), within process variations. In some embodiments, the upper surface 1151U' of the additional seal ring 1151 is protruded from the upper surface 1131U' of the first dielectric layer 1131"-1. The sidewall 1131W' of the first dielectric layer 1131"-1 connected to the upper surface 1131U' may be substantially leveled (or coplanar) with the sidewall 111V' of the first semiconductor substrate 111-1. The first portion 3321' of the insulating encapsulant 332" may extend along the sidewalls (111V' and 1131W') of the first portion 210X and the fourth portion 210W. The first portion 3321' of the insulating encapsulant 332" may be in physical contact with the upper surface 1131U' of the first dielectric layer 1131"-1 and the upper surface 1151U' of the additional seal ring 1151. The semiconductor structure 10F is optionally mounted on the package substrate 20 (see FIG. 2F) using the conductive terminals 142 to form an IC package.

[0072] Embodiments may have one or a combination of the following features and/or advantages. By removing a portion of the bonded structure corresponding to the non-bond areas in the bonded structure prior to the formation of the insulating encapsulant, the risk of delamination propagation in the bonded structure during the formation of the insulating encapsulant may be reduced or eliminated. For example, the step of removing the portion of the bonded structure includes removing a peripheral portion of the first semiconductor die. This may help to reduce the stress applied to the first semiconductor die during the formation of the insulating encapsulant, and the adhesion of the first semiconductor die and semiconductor wafer may be improved. In some embodiments, a portion of the semiconductor wafer directly bonded to the peripheral portion of the first semiconductor die is also removed to ensure the non-bond area does not exist in the bonded structure before forming the insulating encapsulant. In some embodiments,

the first semiconductor die includes one or more ledge(s) which may provide a stepped profile, which can be adhered to the second semiconductor die by the insulating encapsulant to reduce delamination defects in the bonded structure. Accordingly, a semiconductor structure with reduced defects, improved reliability, and improved yield may be achieved.

[0073] Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

[0074] According to some embodiments, a semiconductor structure includes a first semiconductor die, a second semiconductor die underlying and bonded to the first semiconductor die, and an insulating encapsulant disposed over the second semiconductor die. The first semiconductor die includes a semiconductor substrate and an interconnect structure underlying the semiconductor substrate. A maximum lateral dimension of the semiconductor substrate of the first semiconductor die is less than that of the second semiconductor die. The insulating encapsulant at least laterally surrounds the semiconductor substrate of the first semiconductor die.

[0075] According to some embodiments, a semiconductor structure includes a first semiconductor die, a second semiconductor die underlying and bonded to the first semiconductor die, and an insulating encapsulant disposed over the second semiconductor die. The first semiconductor die includes a functional region, a seal ring region surrounding the functional region, and a peripheral region surrounding the seal ring region. The peripheral region of the first semiconductor die is in physical contact with the insulating encapsulant and includes a sidewall substantially aligned with a sidewall of the second semiconductor die.

[0076] According to some embodiments, a manufacturing method of a semiconductor structure includes: performing a bonding process to bond a first semiconductor die to a second semiconductor die, wherein after the bonding process, a first sidewall of the first semiconductor die is substantially leveled with a second sidewall of the second semiconductor die; and forming an insulating encapsulant over the second semiconductor die to laterally surround the first semiconductor die.

[0077] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor structure, comprising:
 - a first semiconductor die comprising a semiconductor substrate and an interconnect structure underlying the semiconductor substrate;
 - a second semiconductor die underlying and bonded to the first semiconductor die, wherein a maximum lateral dimension of the semiconductor substrate of the first semiconductor die is less than that of the second semiconductor die; and
 - an insulating encapsulant disposed over the second semiconductor die and at least laterally surrounding the semiconductor substrate of the first semiconductor die.
2. The semiconductor structure of claim 1, wherein the first semiconductor die comprises a sidewall of the interconnect structure and a sidewall of the semiconductor substrate which is laterally displaced from the sidewall of the interconnect structure.
3. The semiconductor structure of claim 2, wherein the insulating encapsulant extends along the sidewall of the semiconductor substrate and lands on an upper surface of the interconnect structure connected to the sidewall of the interconnect structure.
4. The semiconductor structure of claim 2, wherein an upper surface of the interconnect structure connected to the sidewall of the interconnect structure comprises a surface of a conductive feature which is in direct contact with the insulating encapsulant.
5. The semiconductor structure of claim 1, wherein the first semiconductor die further comprises:
 - a bonding structure underlying the interconnect structure and bonded to the second semiconductor die, wherein the insulating encapsulant is vertically spaced apart from the second semiconductor die by the interconnect structure and the bonding structure.
6. The semiconductor structure of claim 1, wherein:
 - the first semiconductor die further comprises a first bonding structure having a first sidewall,
 - the second semiconductor die comprises a second bonding structure bonded to the first bonding structure and having a second sidewall, and
 - the insulating encapsulant extends along the first and second sidewalls.
7. The semiconductor structure of claim 6, wherein the second semiconductor die further comprises a semiconductor substrate below the second bonding structure, and an outer sidewall of the insulating encapsulant is substantially aligned with a sidewall of the semiconductor substrate of the second semiconductor die.
8. The semiconductor structure of claim 1, wherein the insulating encapsulant comprises:
 - a first portion extending along a first sidewall of the first semiconductor die; and
 - a second portion extending along a second sidewall of the first semiconductor die which is laterally displaced from the first sidewall of the first semiconductor die.
9. The semiconductor structure of claim 8, wherein outer sidewalls of the first and second portions of the insulating encapsulant are substantially aligned with at least a portion of an outer sidewall of the interconnect structure of the first semiconductor die.
10. The semiconductor structure of claim 8, wherein the first and second portions of the insulating encapsulant are

vertically separated from each other by a peripheral region of the first semiconductor die.

11. The semiconductor structure of claim 1, wherein a bonding interface of the first and second semiconductor dies is free of solder material.

12. A semiconductor structure, comprising:

- a first semiconductor die comprising a functional region, a seal ring region surrounding the functional region, and a peripheral region surrounding the seal ring region;

- a second semiconductor die underlying and bonded to the first semiconductor die; and

- an insulating encapsulant disposed over the second semiconductor die, wherein the peripheral region of the first semiconductor die is in physical contact with the insulating encapsulant and comprises a sidewall substantially aligned with a sidewall of the second semiconductor die.

13. The semiconductor structure of claim 12, wherein the sidewall of the peripheral region of the first semiconductor die is substantially aligned with an outer sidewall of the insulating encapsulant, and an upper surface of the peripheral region connected to the sidewall of the peripheral region is in physical contact with the insulating encapsulant.

14. The semiconductor structure of claim 12, wherein the first semiconductor die further comprises a conductive feature disposed in the peripheral region, and a surface of the conductive feature is in physical contact with the insulating encapsulant.

15. The semiconductor structure of claim 12, wherein:

- the second semiconductor die comprises a semiconductor substrate and a bonding structure over the semiconductor substrate and bonded to the first semiconductor die, and

- the sidewall of the peripheral region of the first semiconductor die is substantially aligned with a sidewall of the bonding structure which is laterally displaced from a sidewall of the semiconductor substrate.

16. The semiconductor structure of claim 12, wherein the insulating encapsulant comprises a first portion and a second portion vertically separated from each other by the peripheral region of the first semiconductor die.

17. The semiconductor structure of claim 12, wherein the first portion of the insulating encapsulant is wider than the second portion of the insulating encapsulant.

18. A manufacturing method of a semiconductor structure, comprising:

- performing a bonding process to bond a first semiconductor die to a second semiconductor die, wherein after the bonding process, a first sidewall of the first semiconductor die is substantially leveled with a second sidewall of the second semiconductor die; and

- forming an insulating encapsulant over the second semiconductor die to laterally surround the first semiconductor die.

19. The manufacturing method of claim 18, further comprising:

- partially removing the first semiconductor die to form the first semiconductor die comprising the first sidewall and a third sidewall laterally disposed from the first sidewall, after the bonding process and before forming the insulating encapsulant; and

- performing a singulation process on the insulating encapsulant, the first semiconductor die, and the second

semiconductor die, wherein after the singulation process, the first sidewall of the first semiconductor die is substantially leveled with the second sidewall of the second semiconductor die.

20. The manufacturing method of claim **18**, further comprising:

partially removing a peripheral region of the first semiconductor die and a portion of the second semiconductor die underlying the peripheral region of the first semiconductor die to form a recess on the second semiconductor die, after the bonding process and before forming the insulating encapsulant; and forming the insulating encapsulant on the second semiconductor die to laterally cover the first semiconductor die and fill the recess.

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